T18 메모리 로직 광학을 연결하는 패키징 공동설계

Won Kyoung Choi · Samsung Electronics



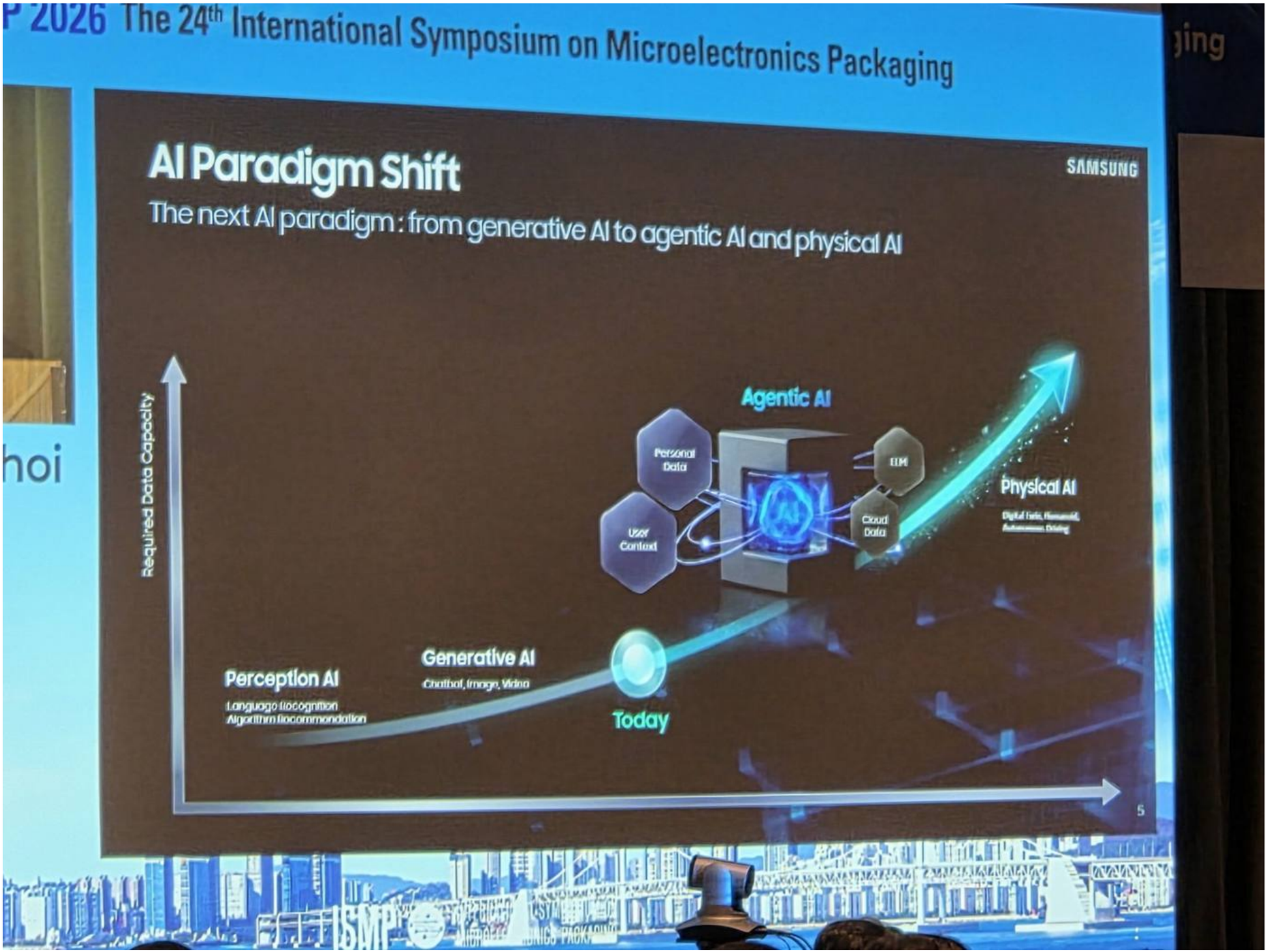
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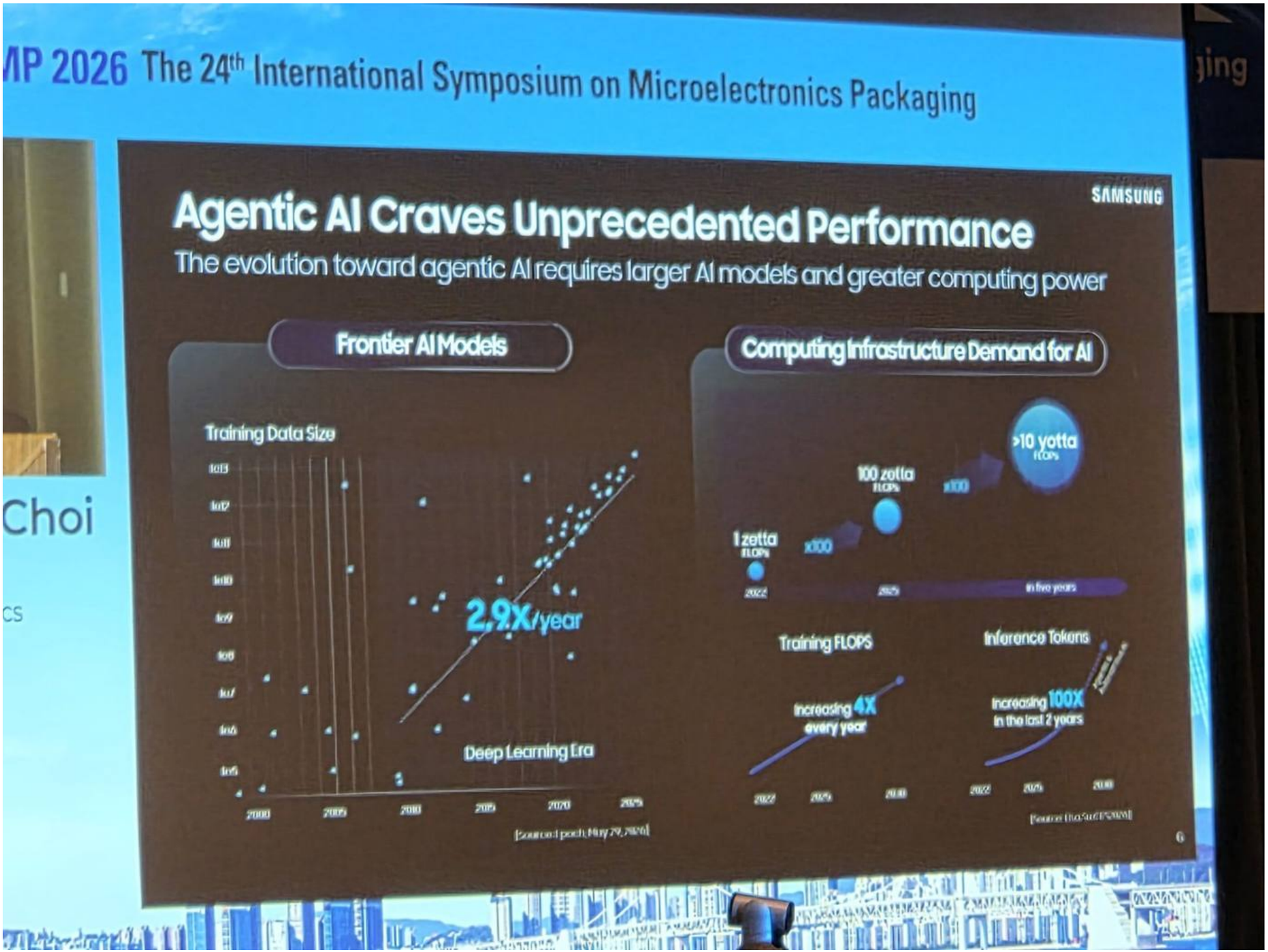
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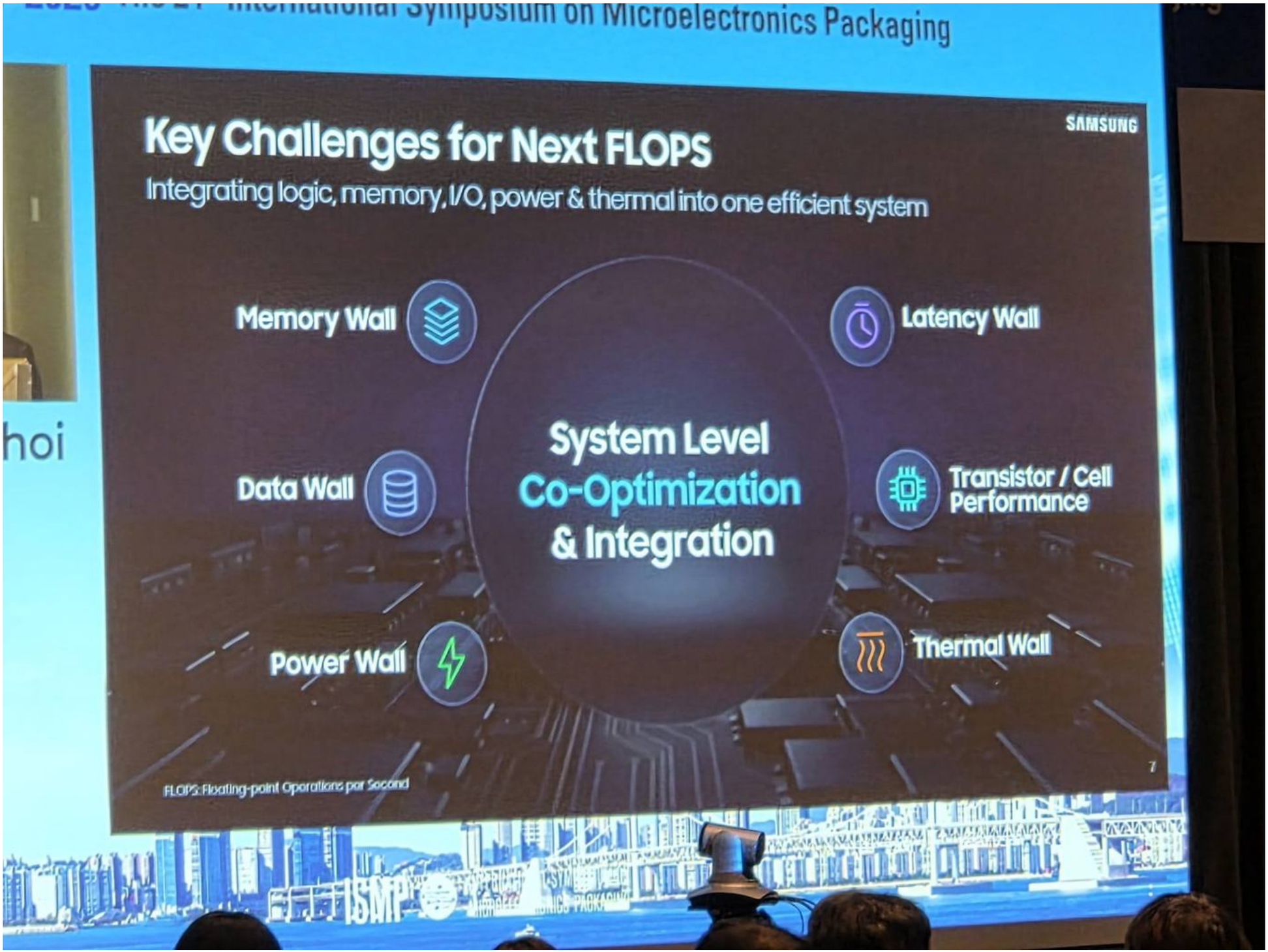
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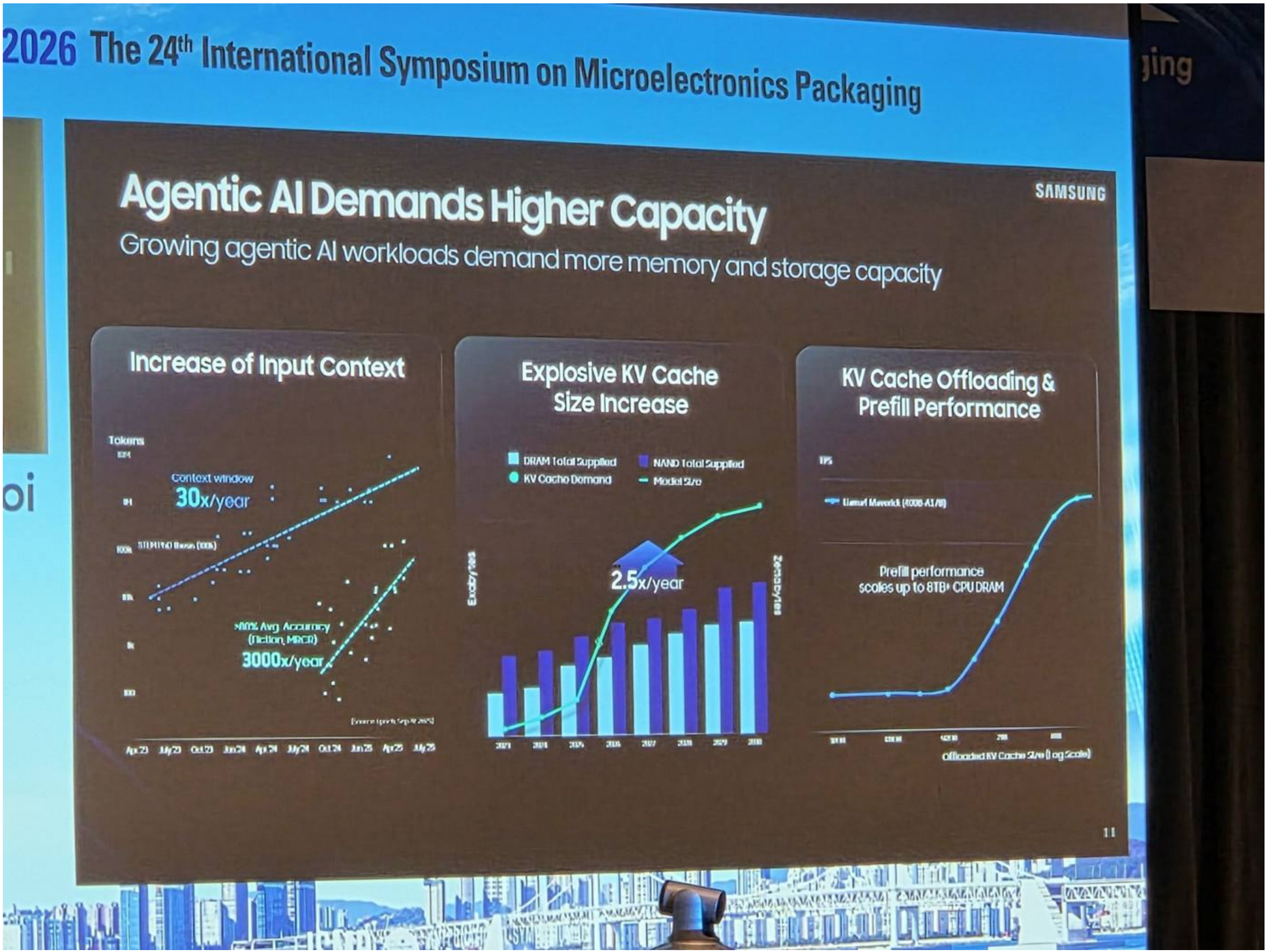
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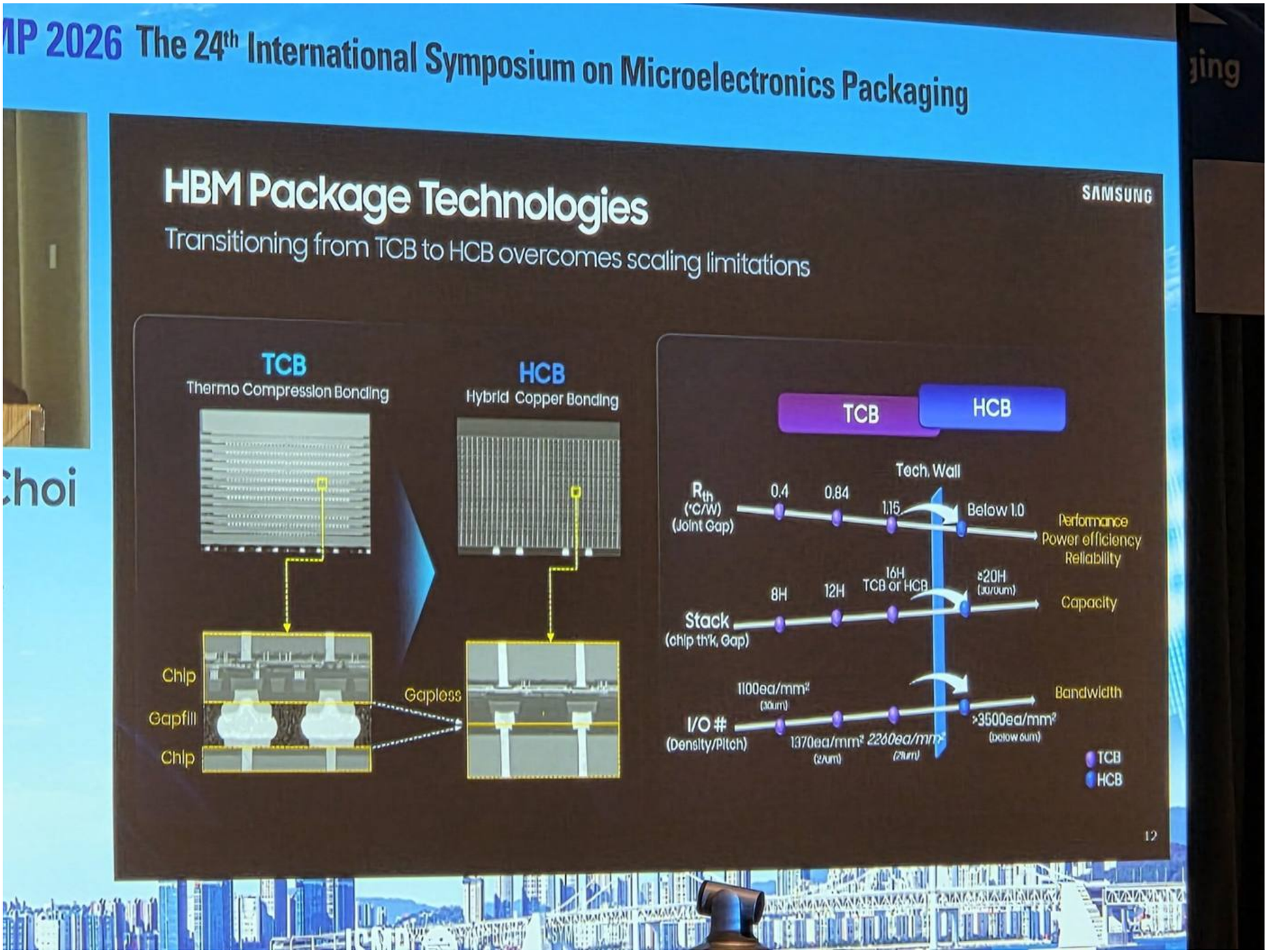
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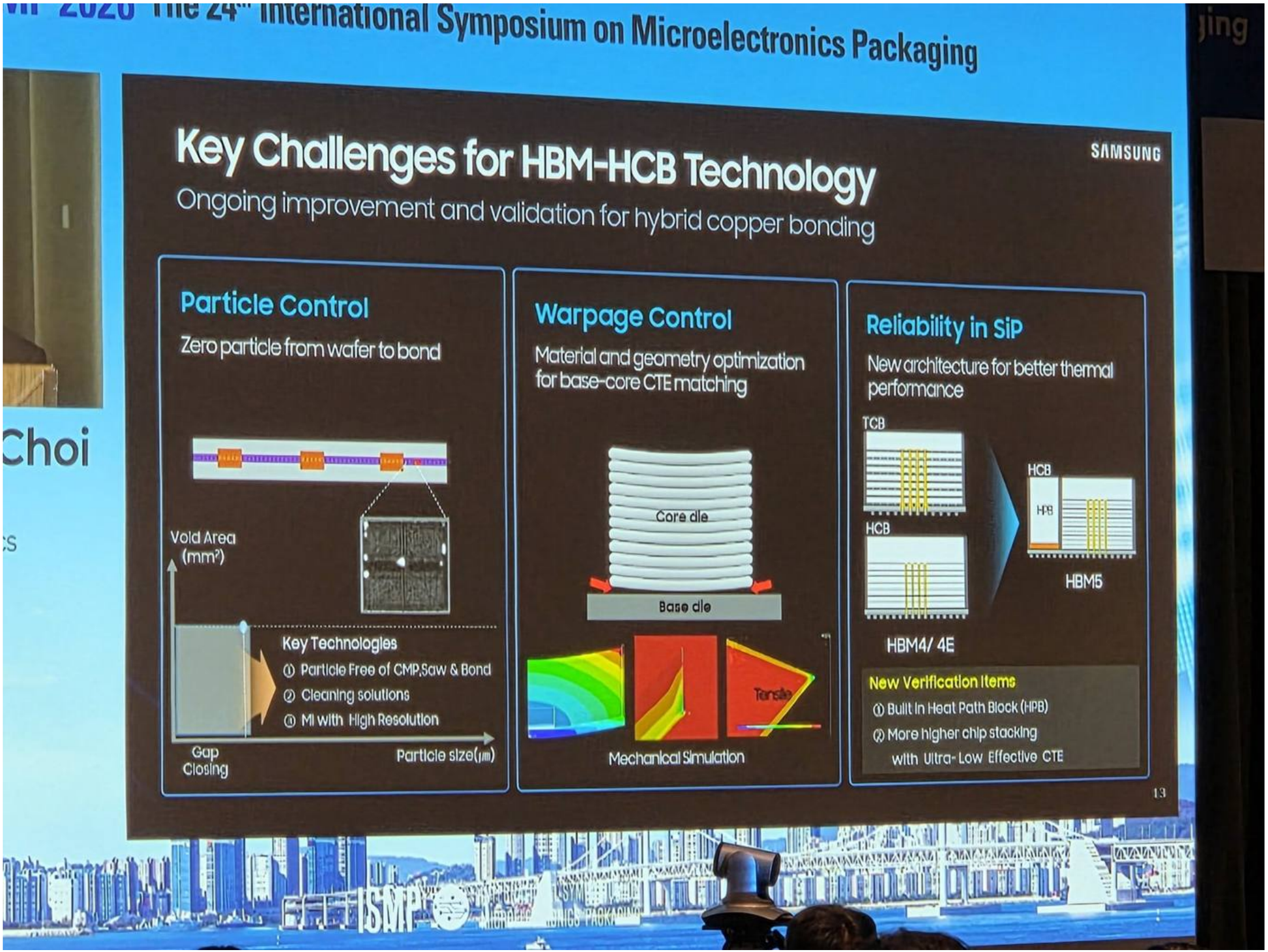
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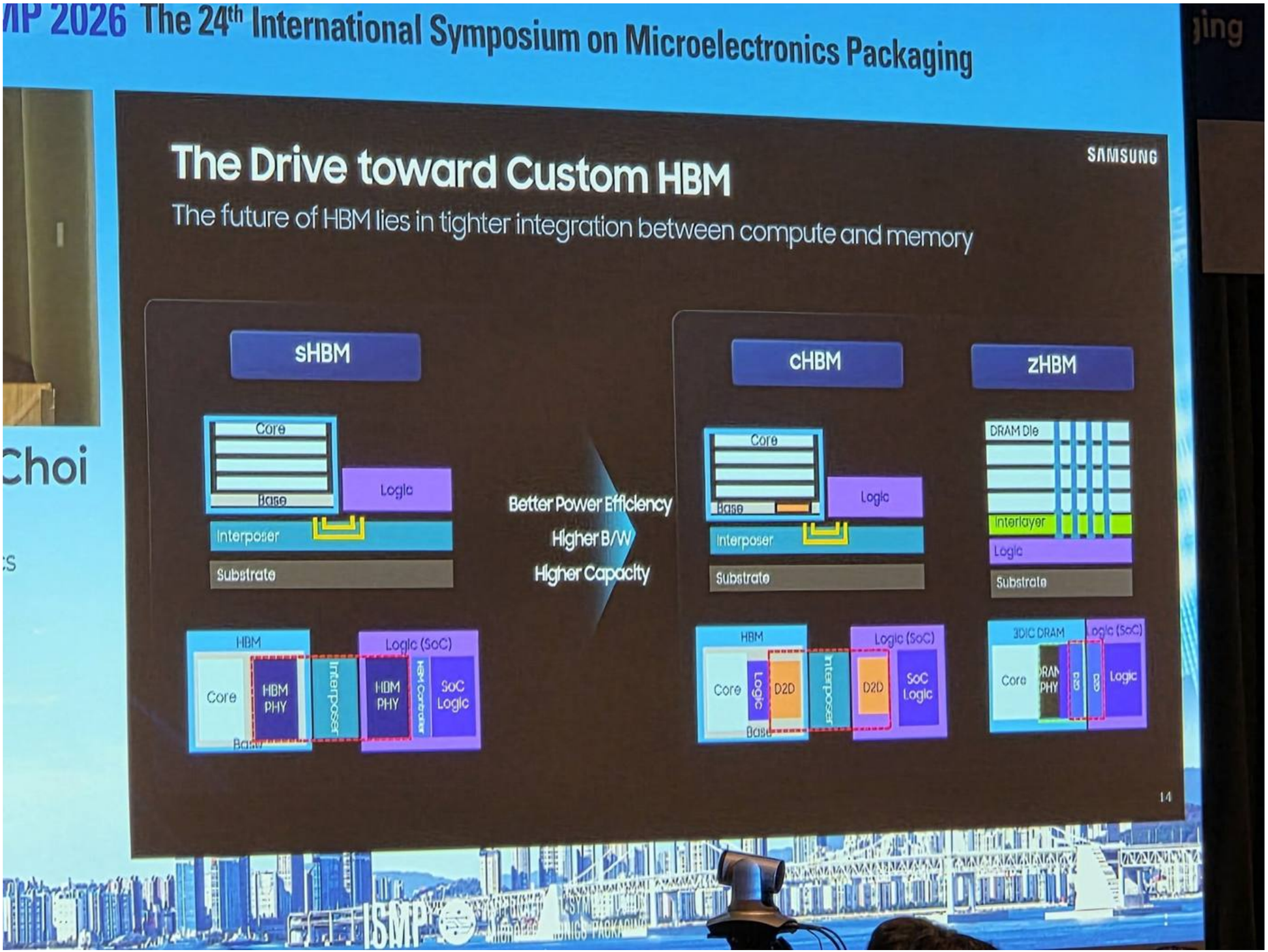
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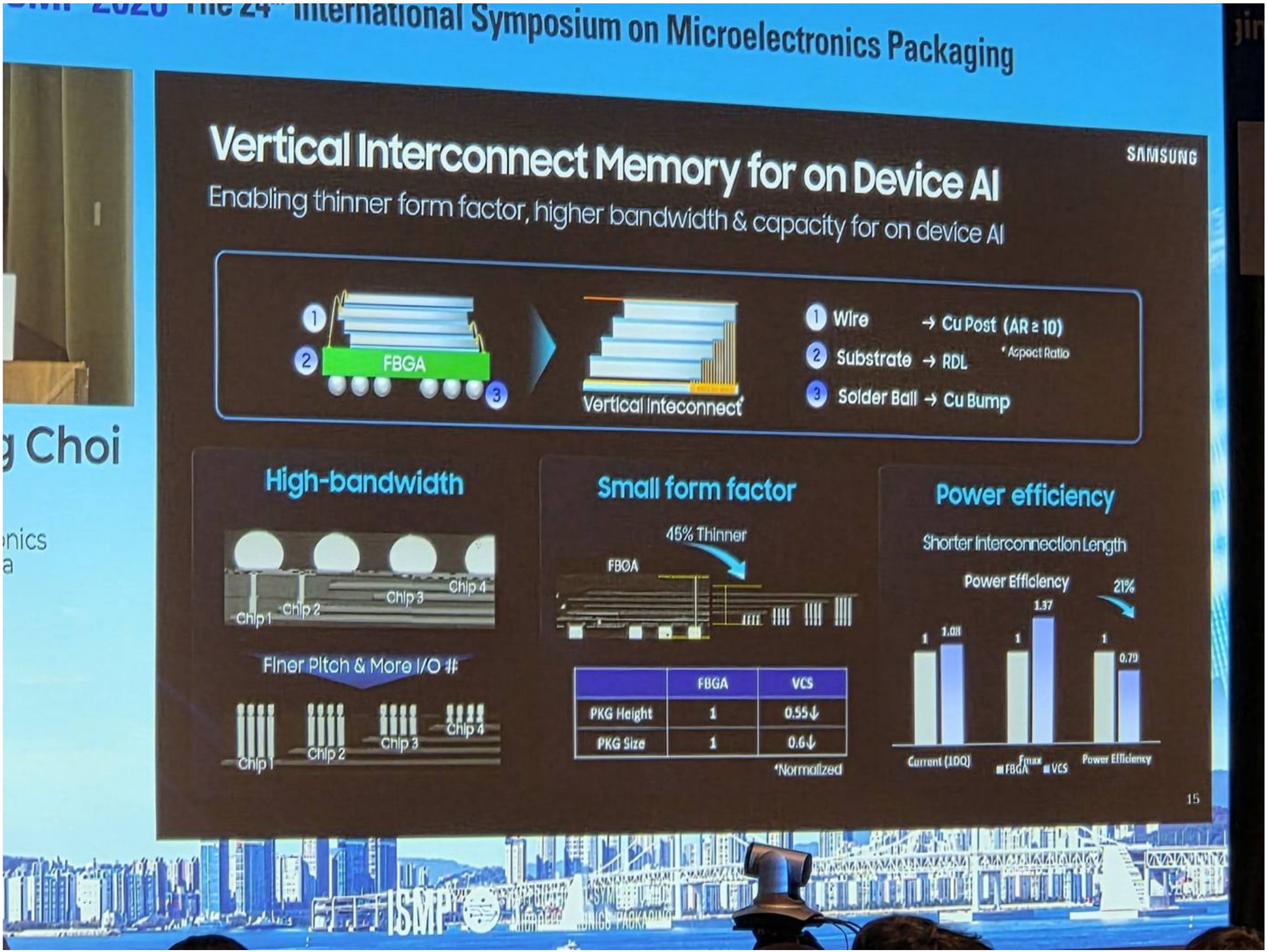
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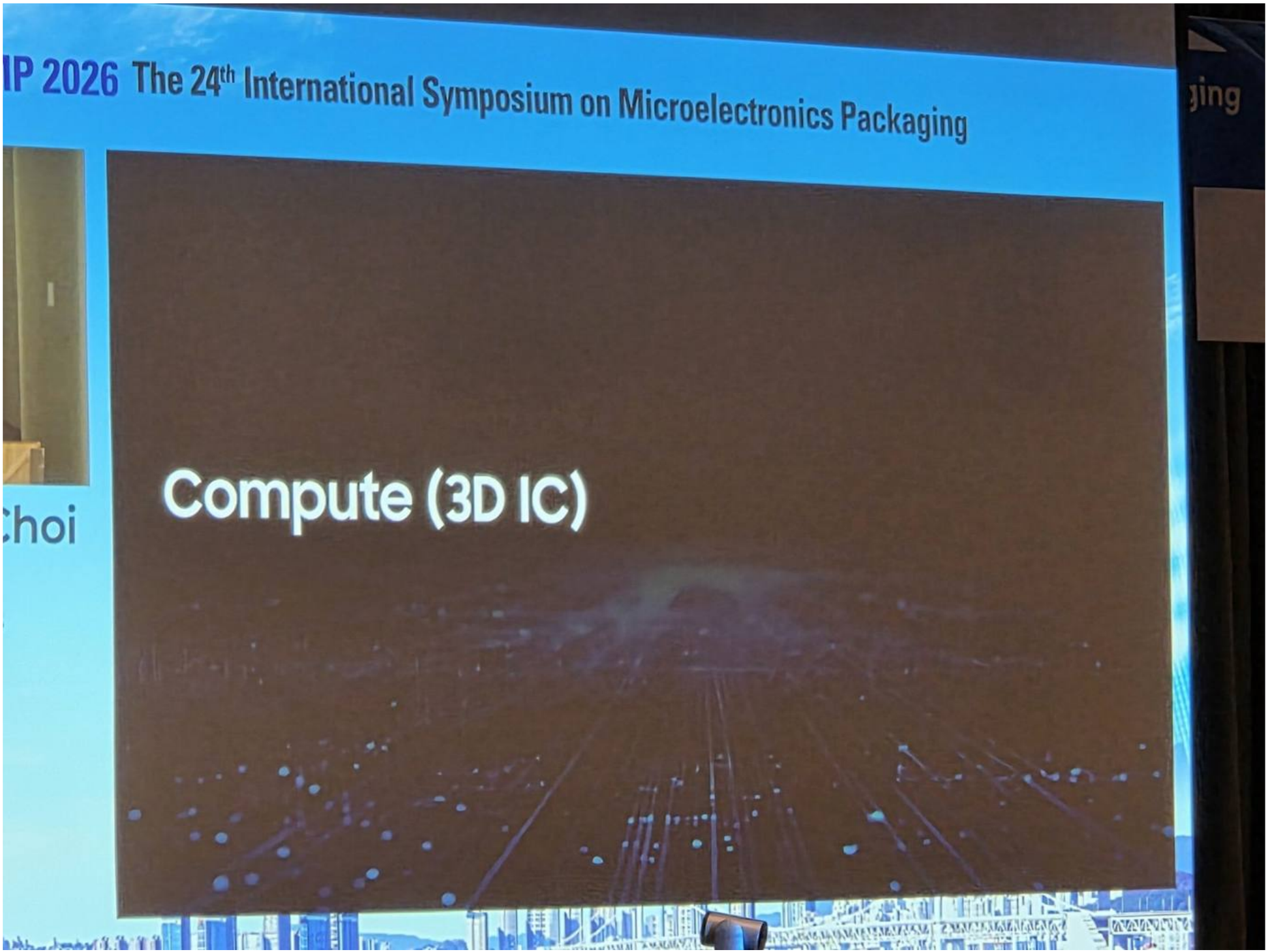
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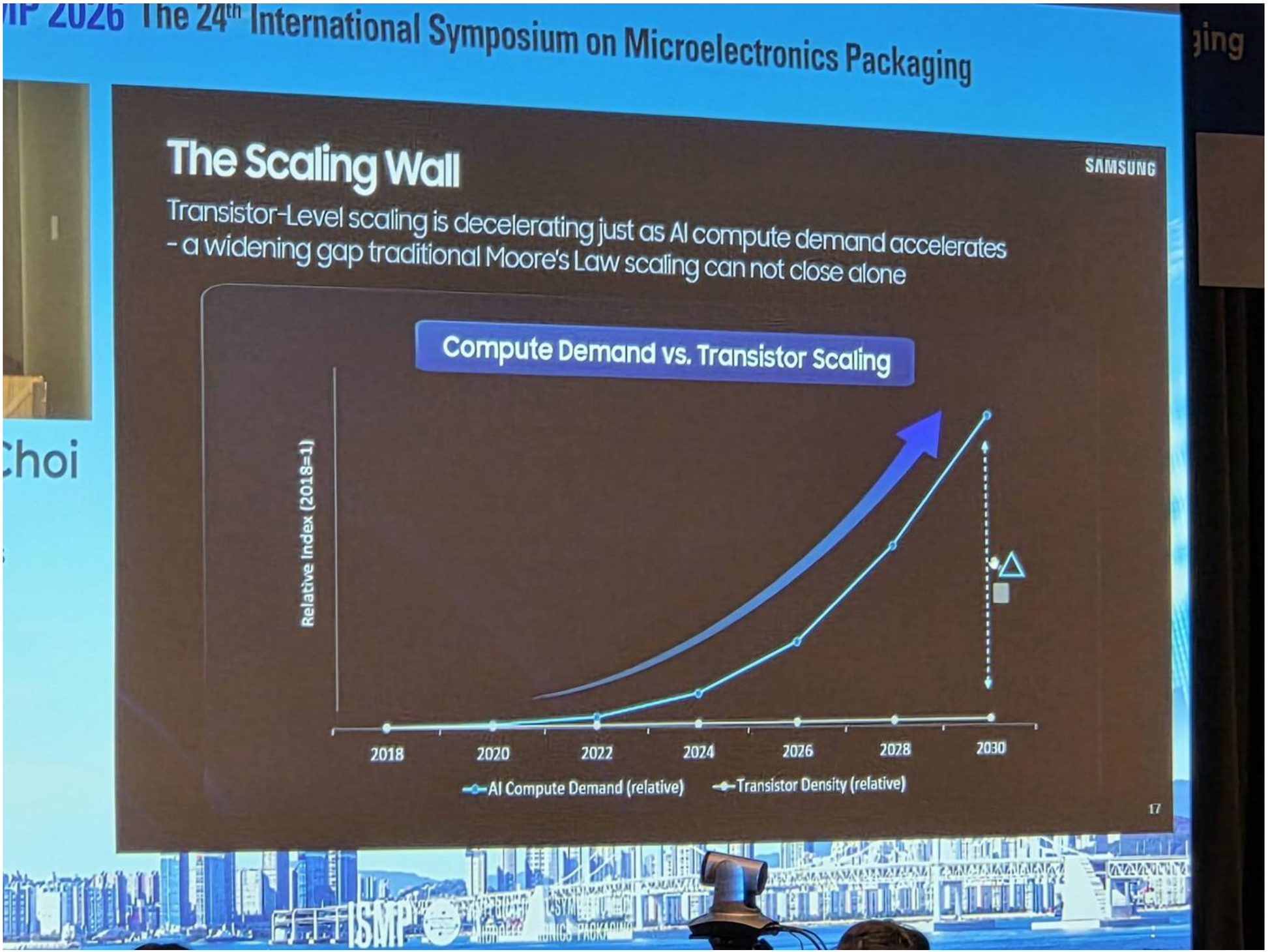
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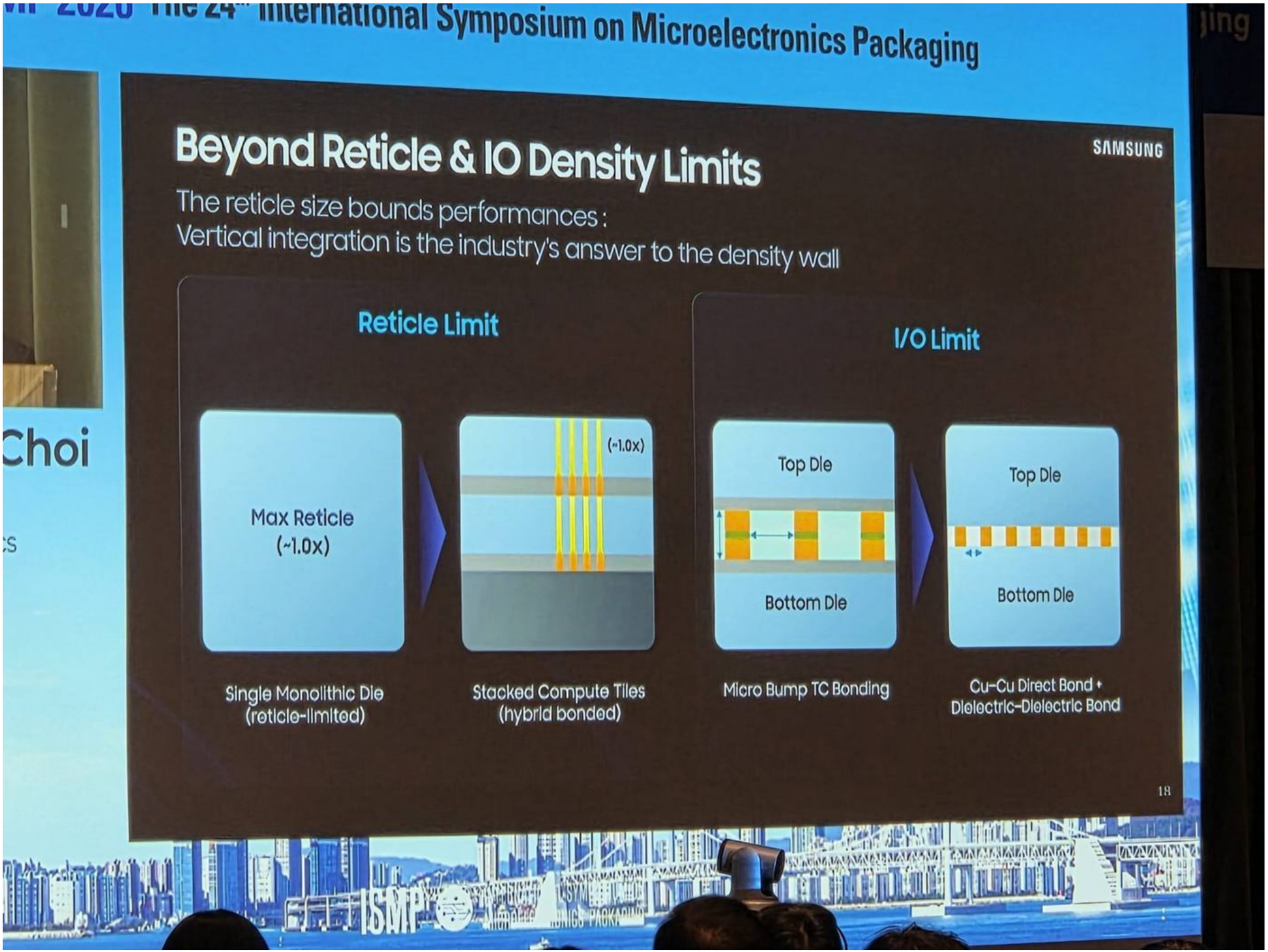
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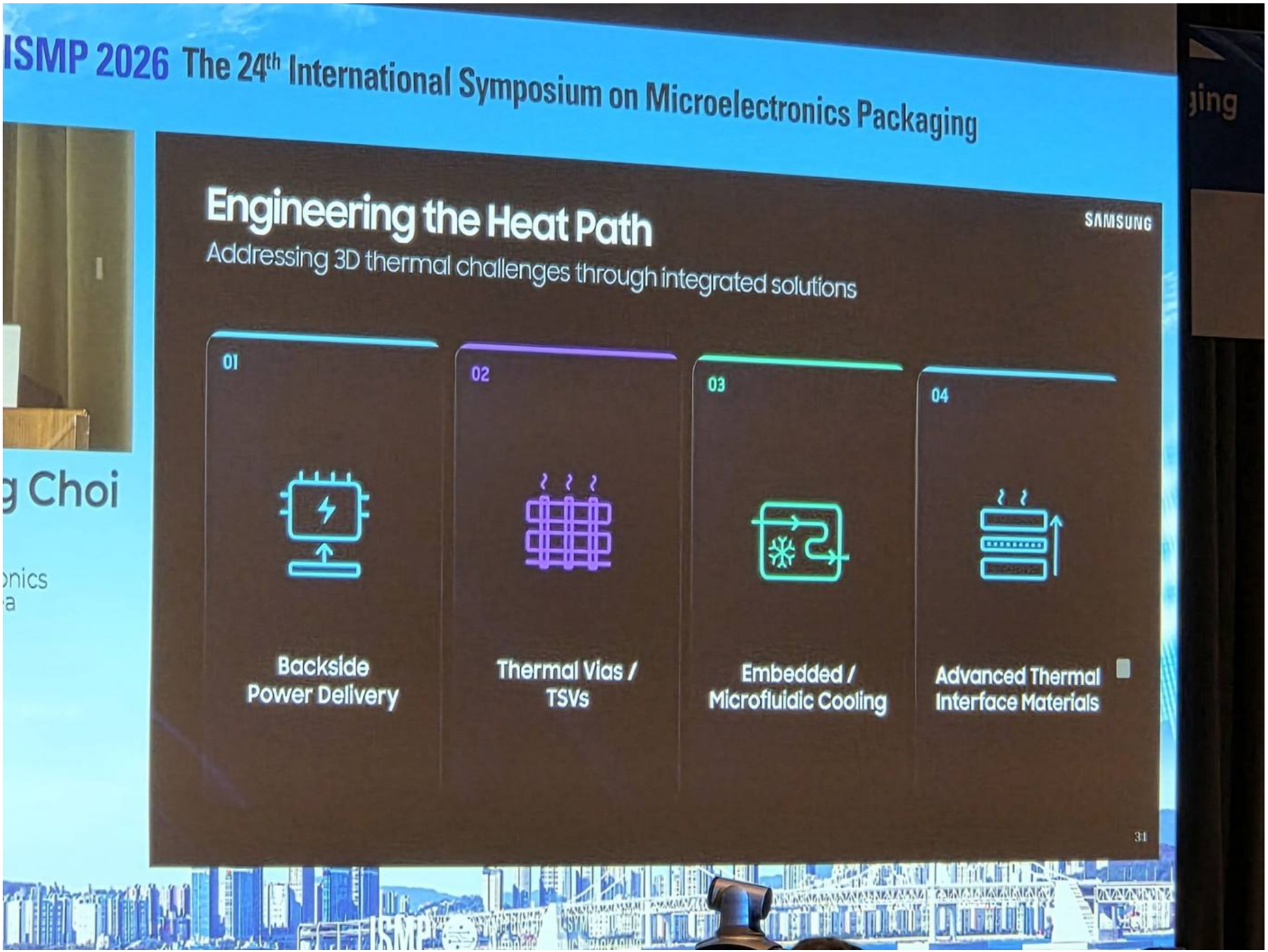
T18 메모리 로직 광학을 연결하는 패키징 공동설계

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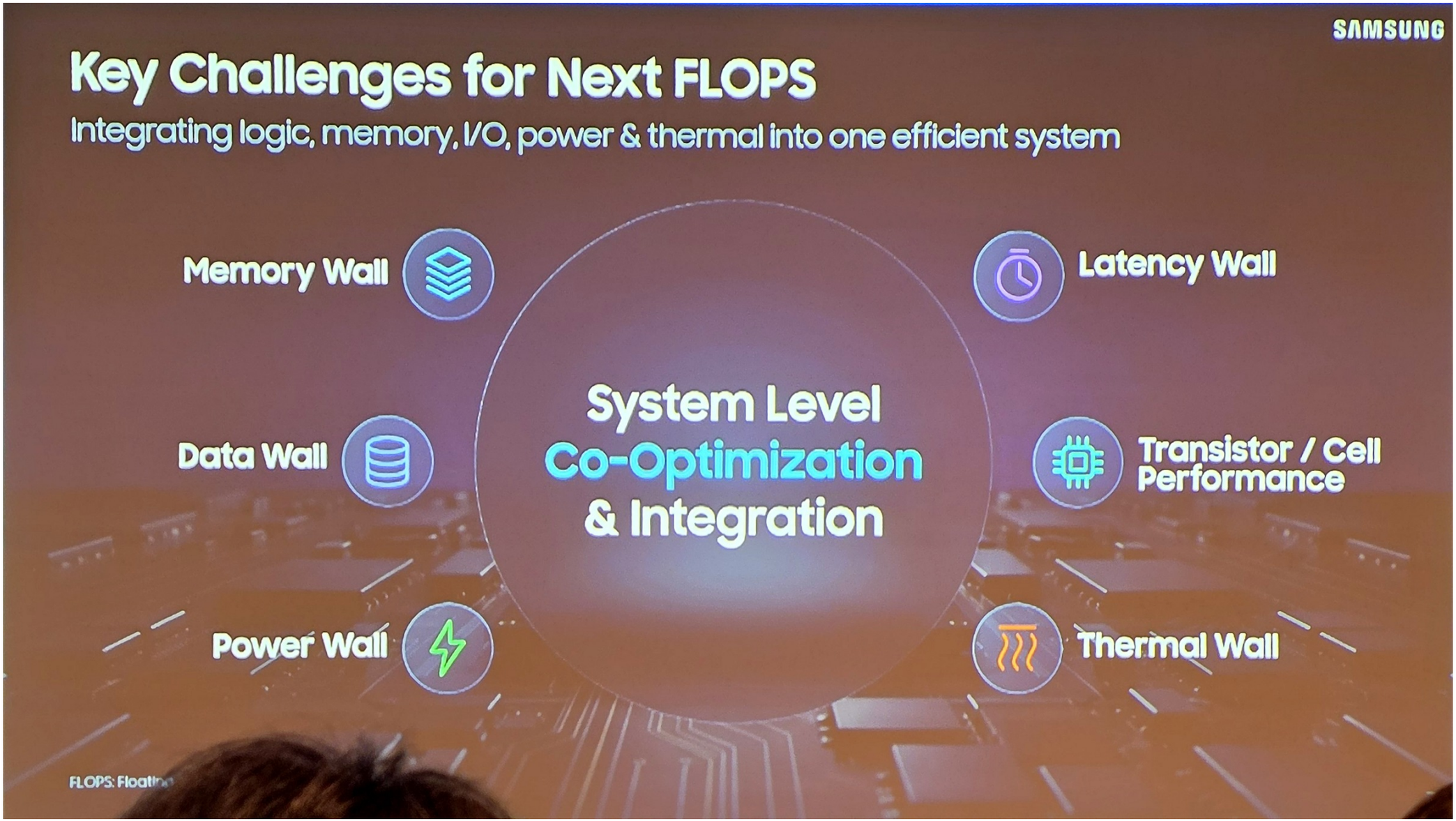
Won Kyoung Choi · Samsung Electronics

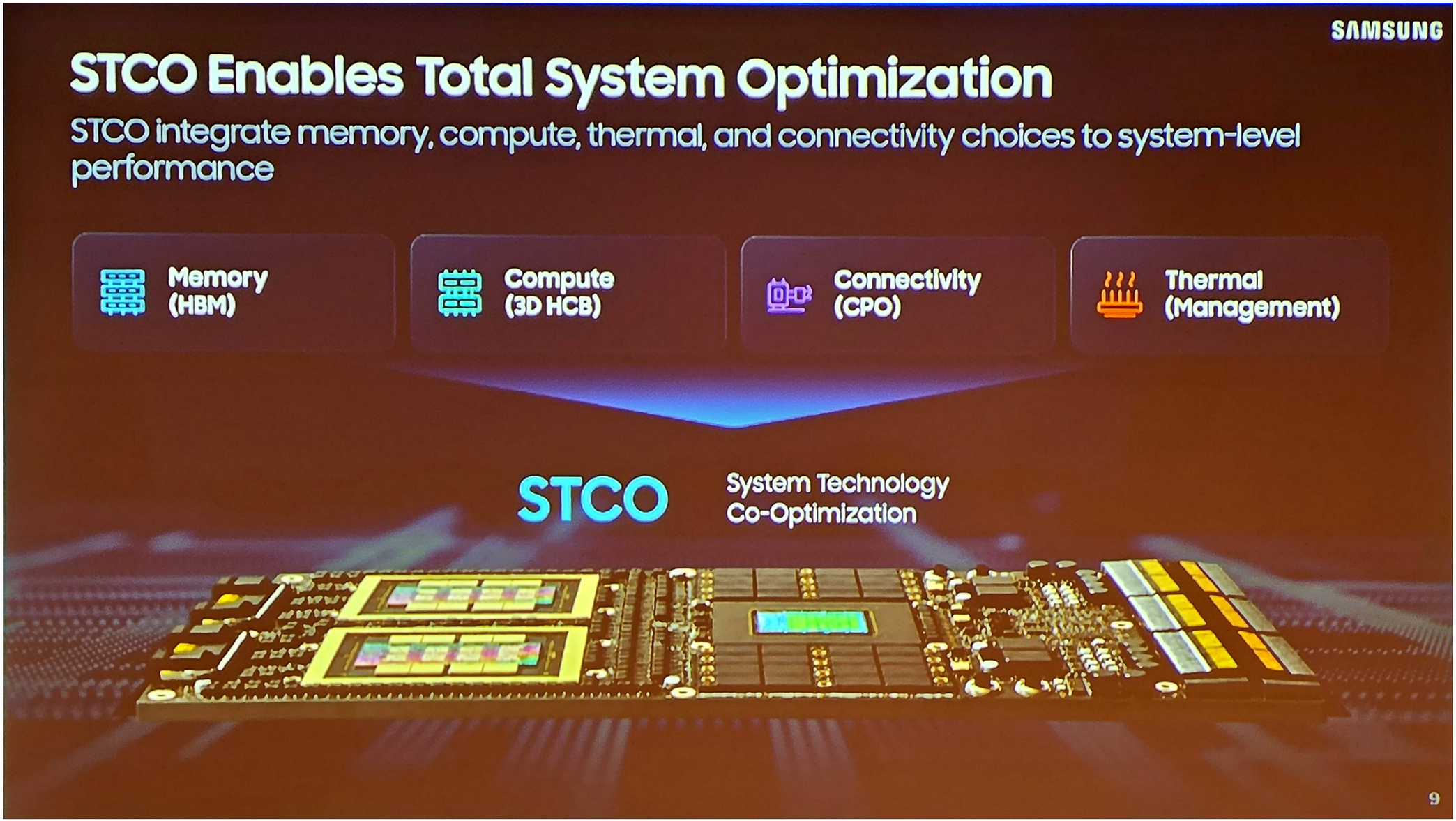


T18 메모리 로직 광학을 연결하는 패키징 공동설계

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SAMSUNG
Key Challenges for Next FLOPS
Integrating logic, memory, I/O, power & thermal into one efficient system
Memory Wall
Latency Wall
Data Wall
System Level
Co-Optimization
& Integration
Transistor / Cell
Performance
Power Wall
Thermal Wall


SAMSUNG
STCO Enables Total System Optimization
STCO integrate memory, compute, thermal, and connectivity choices to system-level performance
Memory (HBM)
Compute (3D HCB)
Connectivity (CPO)
Thermal (Management)
STCO
System Technology Co-Optimization
9

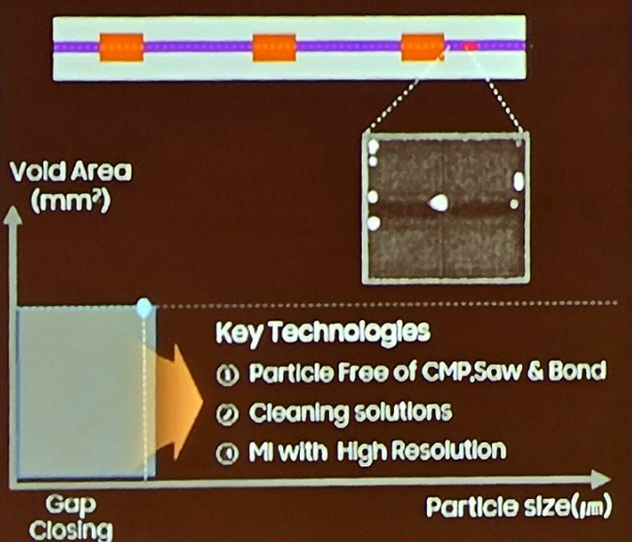
SAMSUNG

Key Challenges for HBM-HCB Technology

Ongoing improvement and validation for hybrid copper bonding

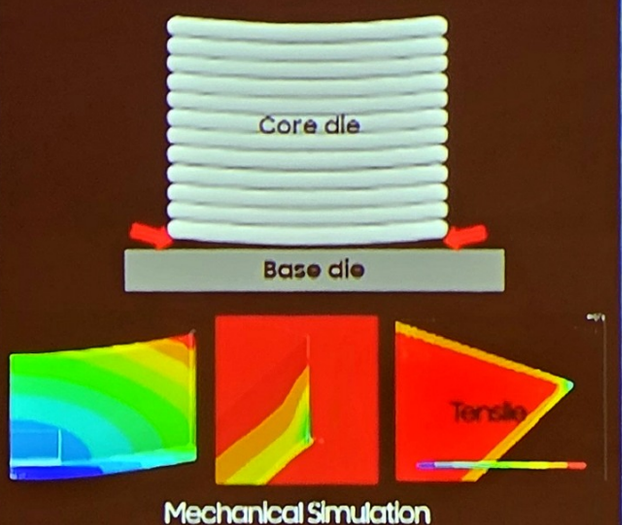
Particle Control

Zero particle from wafer to bond



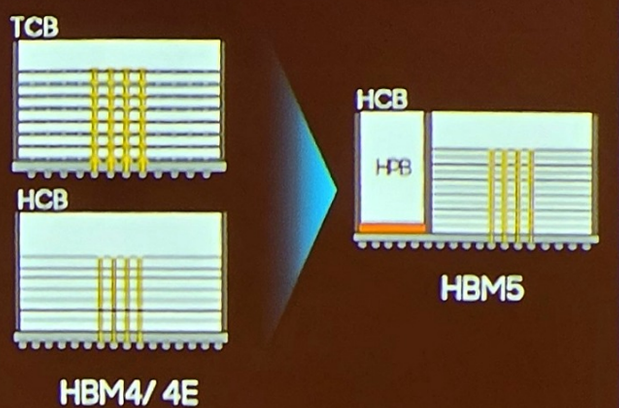
Warpage Control

Material and geometry optimization for base-core CTE matching



Reliability in SiP

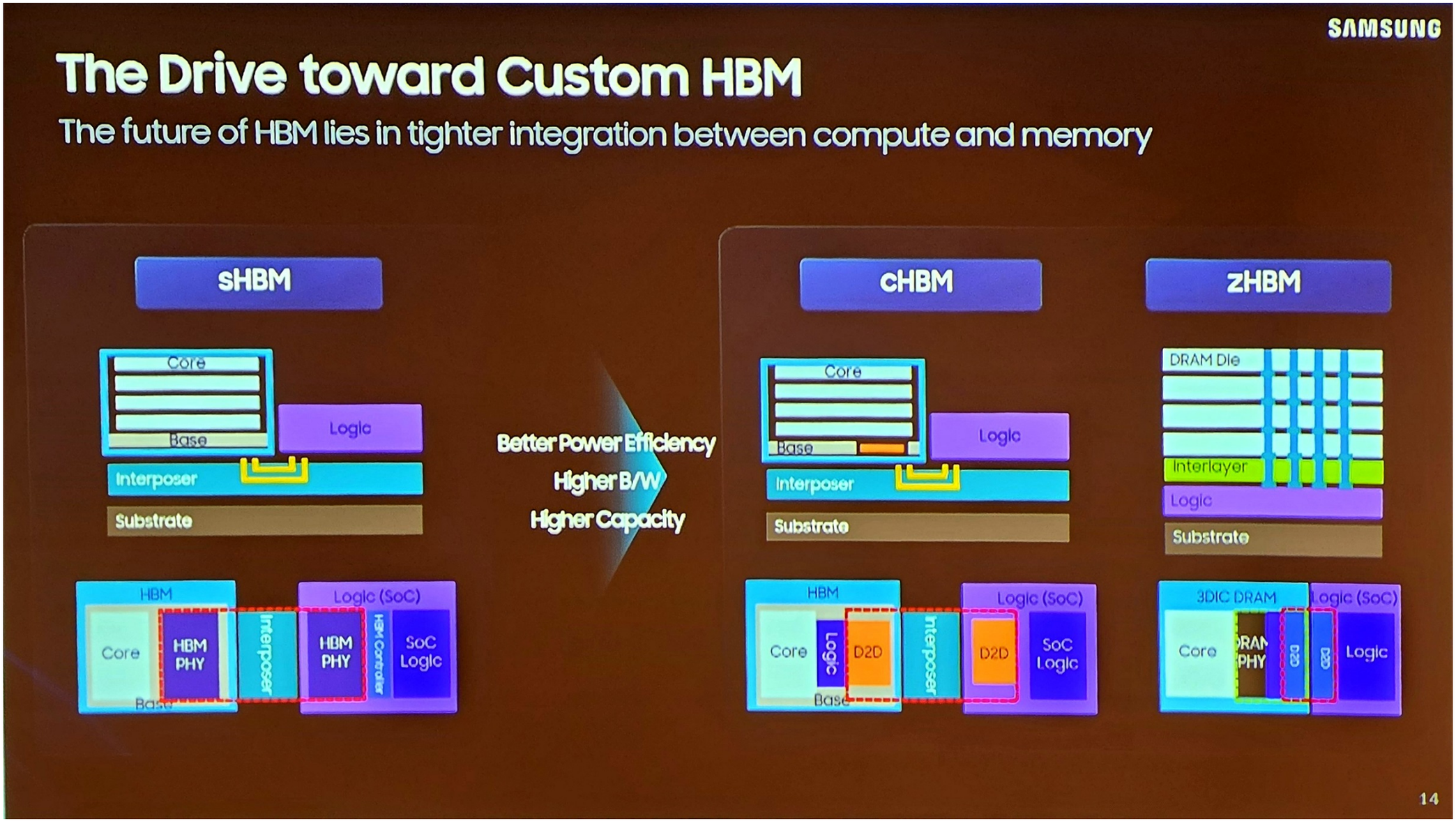
New architecture for better thermal performance

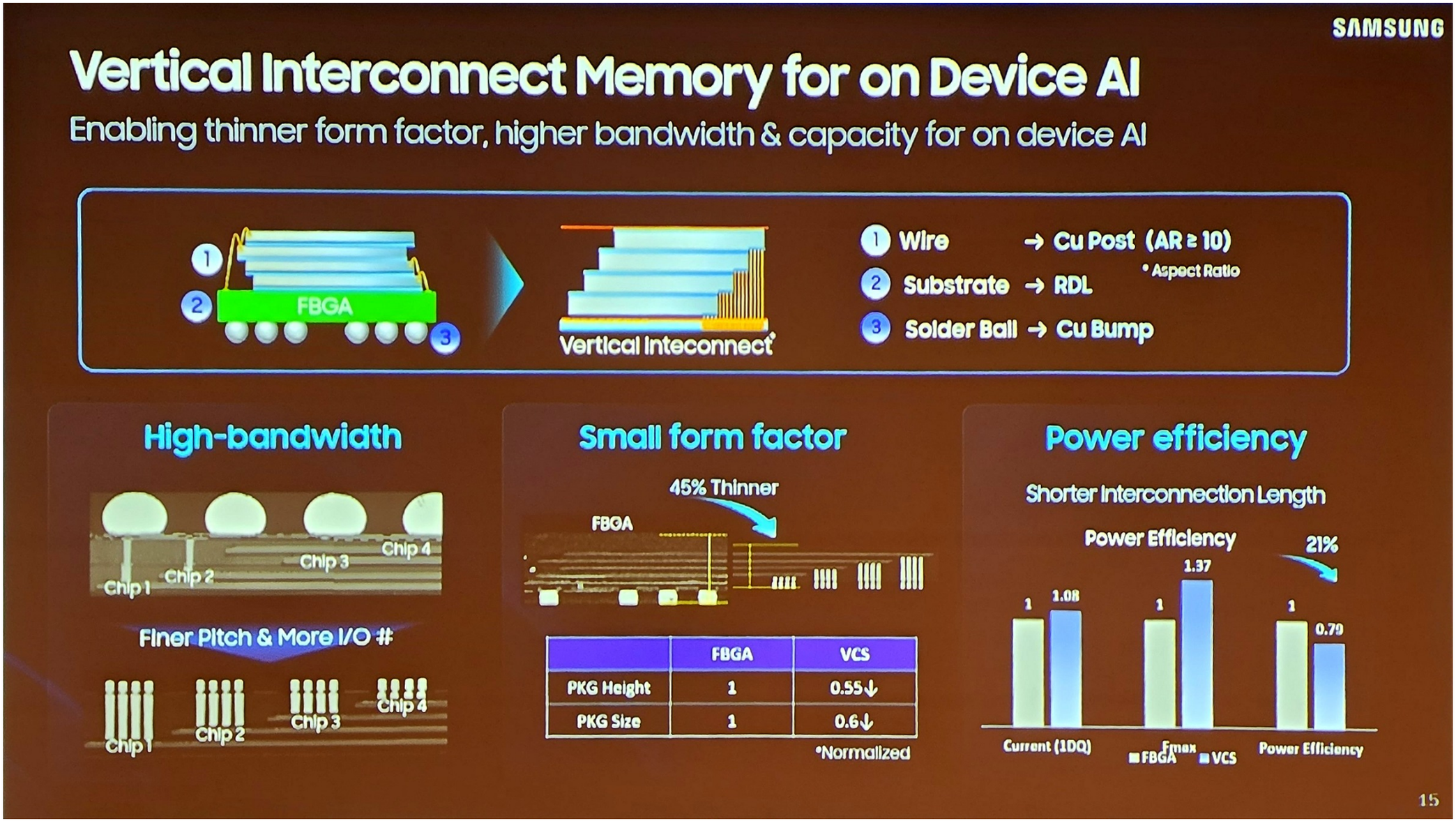


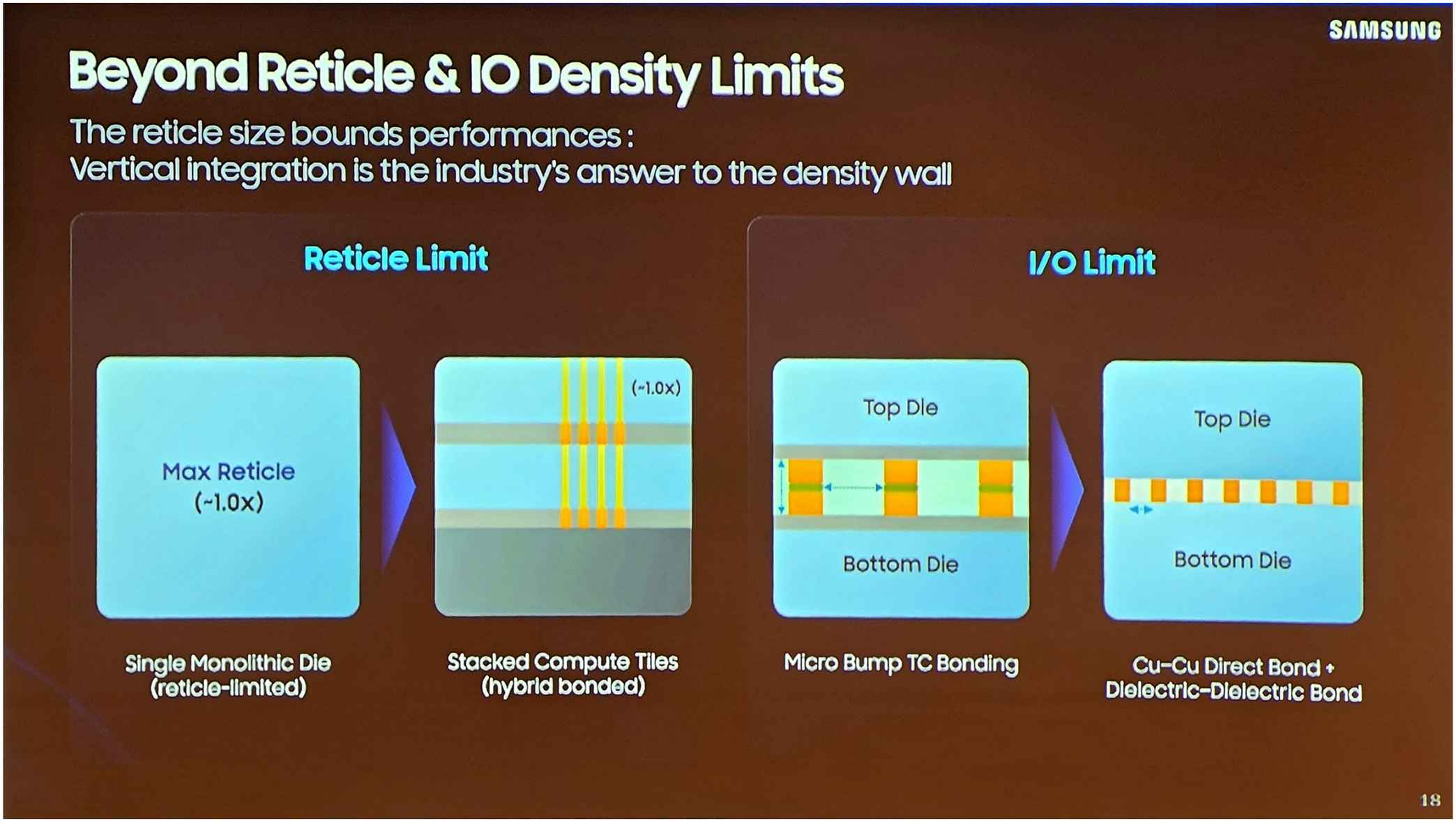
New Verification Items

① Built In Heat Path Block (HPB)
② More higher chip stacking with Ultra-Low Effective CTE

13


SAMSUNG
The Drive toward Custom HBM
The future of HBM lies in tighter integration between compute and memory
sHBM
cHBM
zHBM
Core
Base
Logic
Interposer
Substrate
Better Power Efficiency
Higher B/W
Higher Capacity
DRAM Die
Interlayer
HBM
Logic (SoC)
HBM PHY
SoC Logic
D2D
3DIC DRAM
14


SAMSUNG
Vertical Interconnect Memory for on Device AI
Enabling thinner form factor, higher bandwidth & capacity for on device AI
1
2
FBGA
3
Vertical Inteconnect
1 Wire → Cu Post (AR ≥ 10)
2 Substrate → RDL
3 Solder Ball → Cu Bump
* Aspect Ratio
High-bandwidth
Chip 1
Chip 2
Chip 3
Chip 4
Finer Pitch & More I/O #
Small form factor
45% Thinner
FBGA
FBGA VCS
PKG Height 1 0.55↓
PKG Size 1 0.6↓
*Normalized
Power efficiency
Shorter Interconnection Length
Power Efficiency
21%
1
1.08
1
1.37
1
0.79
Current (1DQ)
Fmax
FBGA VCS
Power Efficiency
15

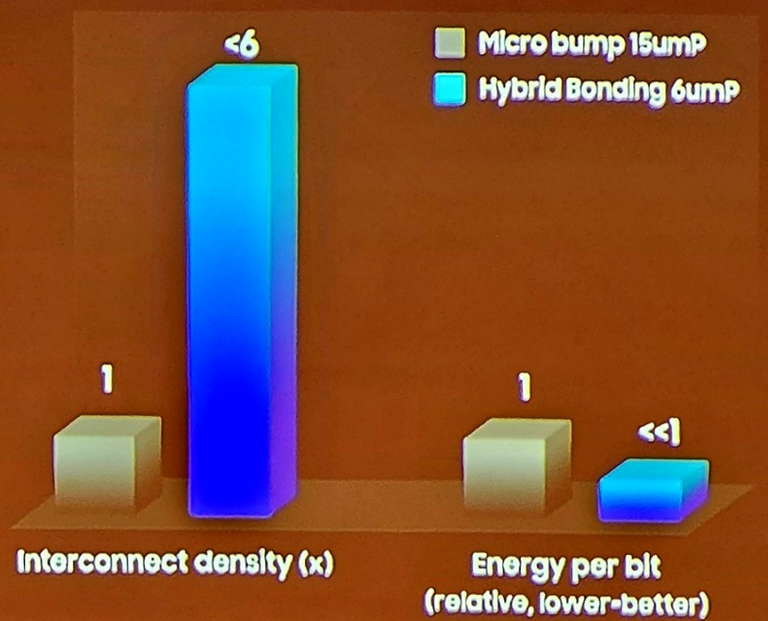

SAMSUNG
Beyond Reticle & IO Density Limits
The reticle size bounds performances :
Vertical integration is the industry's answer to the density wall
Reticle Limit
I/O Limit
Max Reticle
(~1.0x)
(~1.0x)
Top Die
Bottom Die
Top Die
Bottom Die
Single Monolithic Die
(reticle-limited)
Stacked Compute Tiles
(hybrid bonded)
Micro Bump TC Bonding
Cu-Cu Direct Bond +
Dielectric-Dielectric Bond
18

SAMSUNG

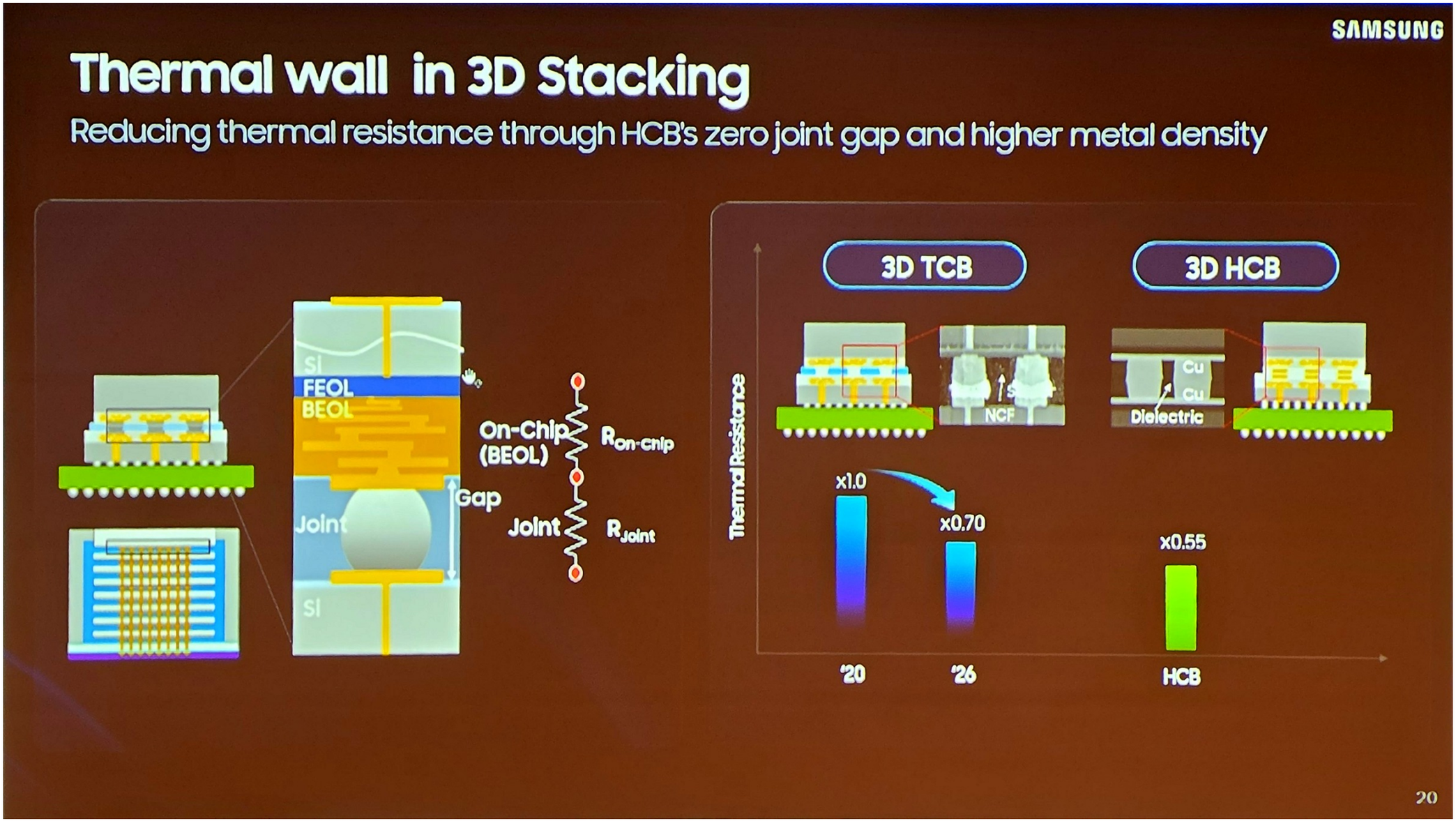
Interconnect Pitch Scaling

Hybrid bonding pushes interconnect pitch below 15um enabling higher interconnect density

	Interconnect pitch	Typical use
Micro bump	~25 – 45 µm	HBM, 2.xD
Fine pitch bump	~15 – 25 µm	3D logic, HBM Advanced 2.xD
Hybrid bonding (3D)	< 15 µm	3D logic, HBM ZHBM



19


SAMSUNG
Thermal wall in 3D Stacking
Reducing thermal resistance through HCB's zero joint gap and higher metal density
Si
FEOL
BEOL
Gap
Joint
Si
On-Chip (BEOL)
$R_{on\text{-}chip}$
Joint
R_{Joint}
3D TCB
3D HCB
NCF
Cu
Cu
Dielectric
Thermal Resistance
x1.0
x0.70
x0.55
'20
'26
HCB
20

SAMSUNG

3D Logic Technologies

3D integration of multi-chiplets in 3D packages for higher compute

Vertical Stacking (Die Disaggregation)

Up to 40% Reduction vs Monolithic Die

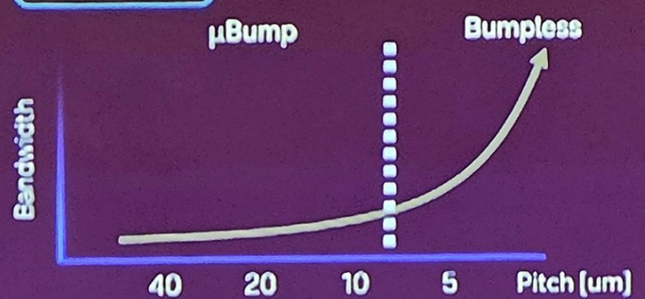
Foot Print

Smaller Footprint (Die placement area)

Logic/Memory

Logic die

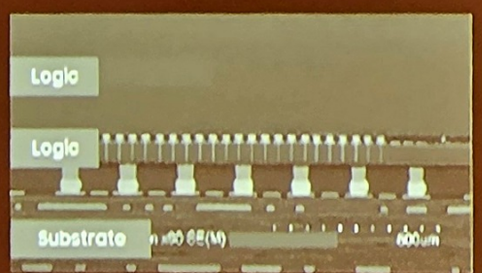
3D



3D TCB



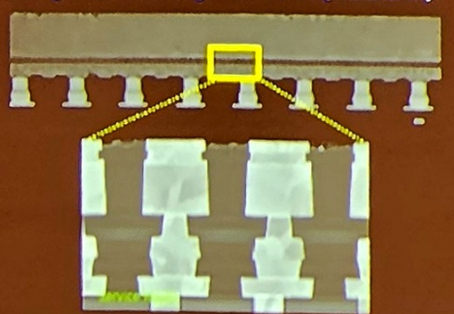
- Massive& Proven tech. & Infra
- Cost Efficient Platform
- 25um Pitch, Back side RDL(2L)
- Reliability R2000 (JEDEC)



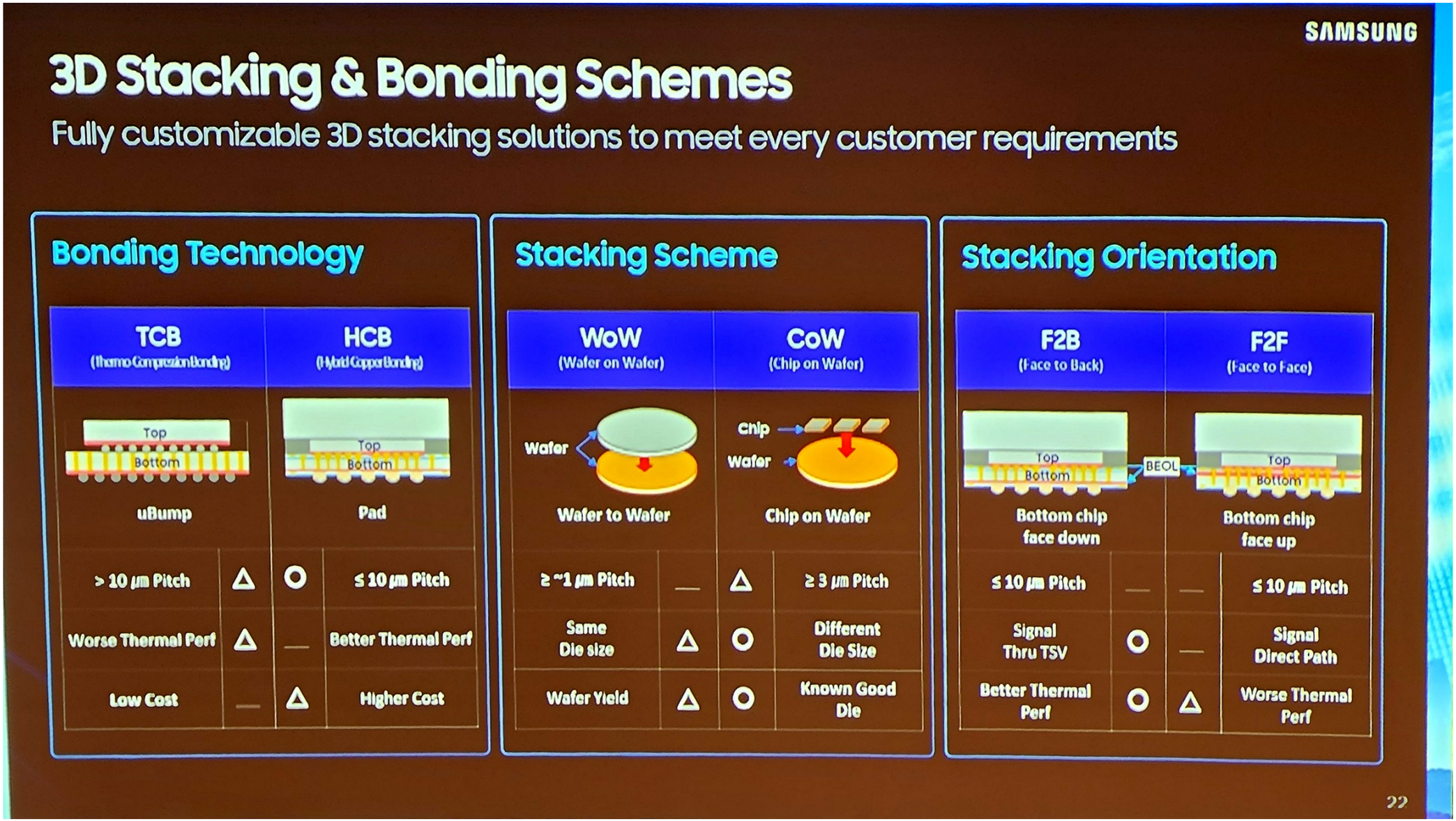
3D HCB



- HBM & 3D Logic R&D Path
- Bandwidth & Thermal Gains
- SF2P 6um Pitch, F2F / F2B
- Early Reliability R2000 (JEDEC)



21

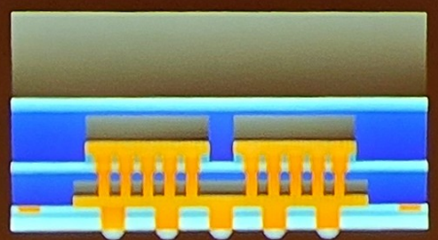

SAMSUNG
3D Stacking & Bonding Schemes
Fully customizable 3D stacking solutions to meet every customer requirements
Bonding Technology
TCB
(Thermo Compression Bonding)
HCB
(Hybrid Copper Bonding)
Top
Bottom
uBump
Pad
> 10 μm Pitch
≤ 10 μm Pitch
Worse Thermal Perf
Better Thermal Perf
Low Cost
Higher Cost
Stacking Scheme
WoW
(Wafer on Wafer)
CoW
(Chip on Wafer)
Wafer
Chip
Wafer to Wafer
Chip on Wafer
≥ ~1 μm Pitch
≥ 3 μm Pitch
Same Die size
Different Die Size
Wafer Yield
Known Good Die
Stacking Orientation
F2B
(Face to Back)
F2F
(Face to Face)
BEOL
Bottom chip face down
Bottom chip face up
≤ 10 μm Pitch
≤ 10 μm Pitch
Signal Thru TSV
Signal Direct Path
Better Thermal Perf
Worse Thermal Perf
22

SAMSUNG

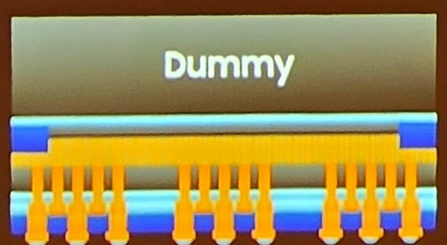
Core Technologies for 3D Logic

Advanced 3D IC integration through core technologies for fine-pitch hybrid bonding and max die size

3D IC Architectures

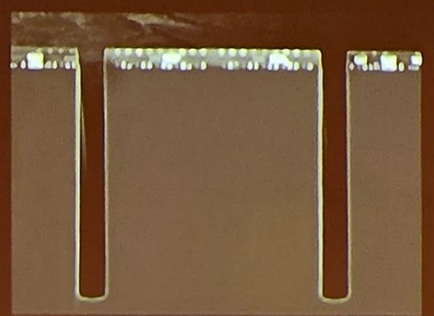


F2B (Face to Back)



F2F (Face to Face)

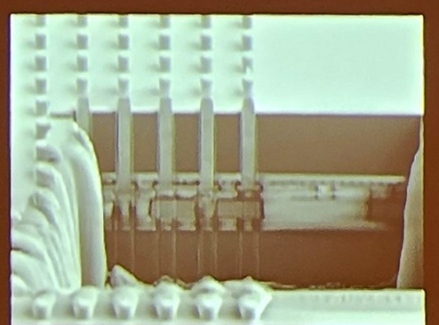
Key Technologies



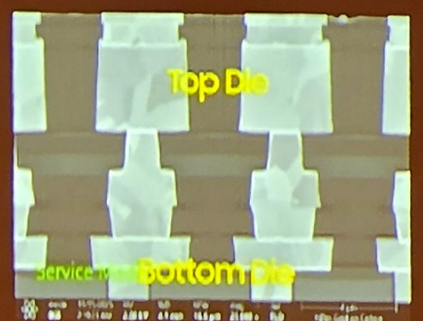
[Plasma dicing]



[Gapfill/ Planarization]



[TSV Reveal]



[HCB CoW]
(Top on bottom Die)

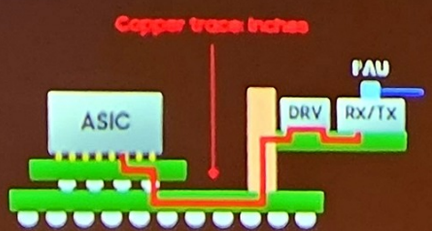
23

SAMSUNG

Si-Photonics for Optical Connect

PO to CPO advancement for better signal, bandwidth and power efficiency

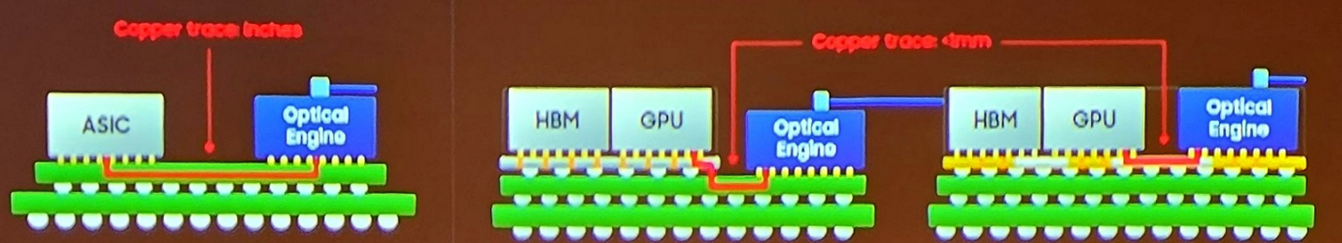
Pluggable Optics (PO)



Data center to Data center

Bandwidth: ~1.6T
Copper trace: inches

Co-Packaged Optics (CPO)



Switch CPO

Network / Data transmission

Bandwidth: ~12.8T
Copper trace: millimeters

GPU CPO

AI / Parallel Computation

Bandwidth: 12.8T ~
Copper trace: <1mm

From PO to CPO: Shorter Electrical Path, Higher Bandwidth, Lower Power Consumption

25

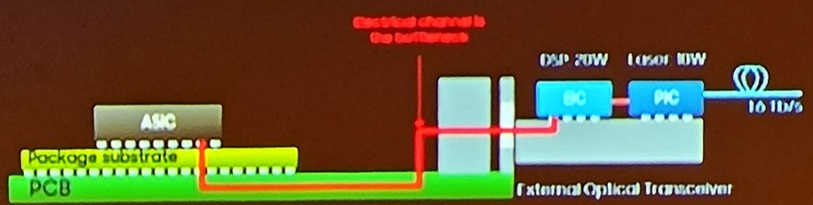
Samsung Electronics · Won Kyoung Choi

SAMSUNG

Overcoming Electrical Limits via CPO

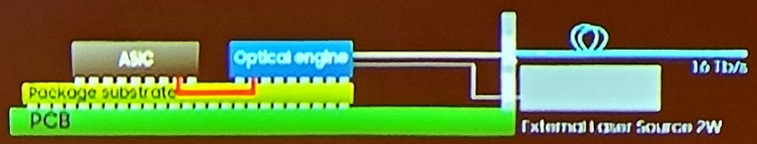
Shifting electrical to optical interconnect for maximum system efficiency and scalability

Traditional Pluggable · 30W



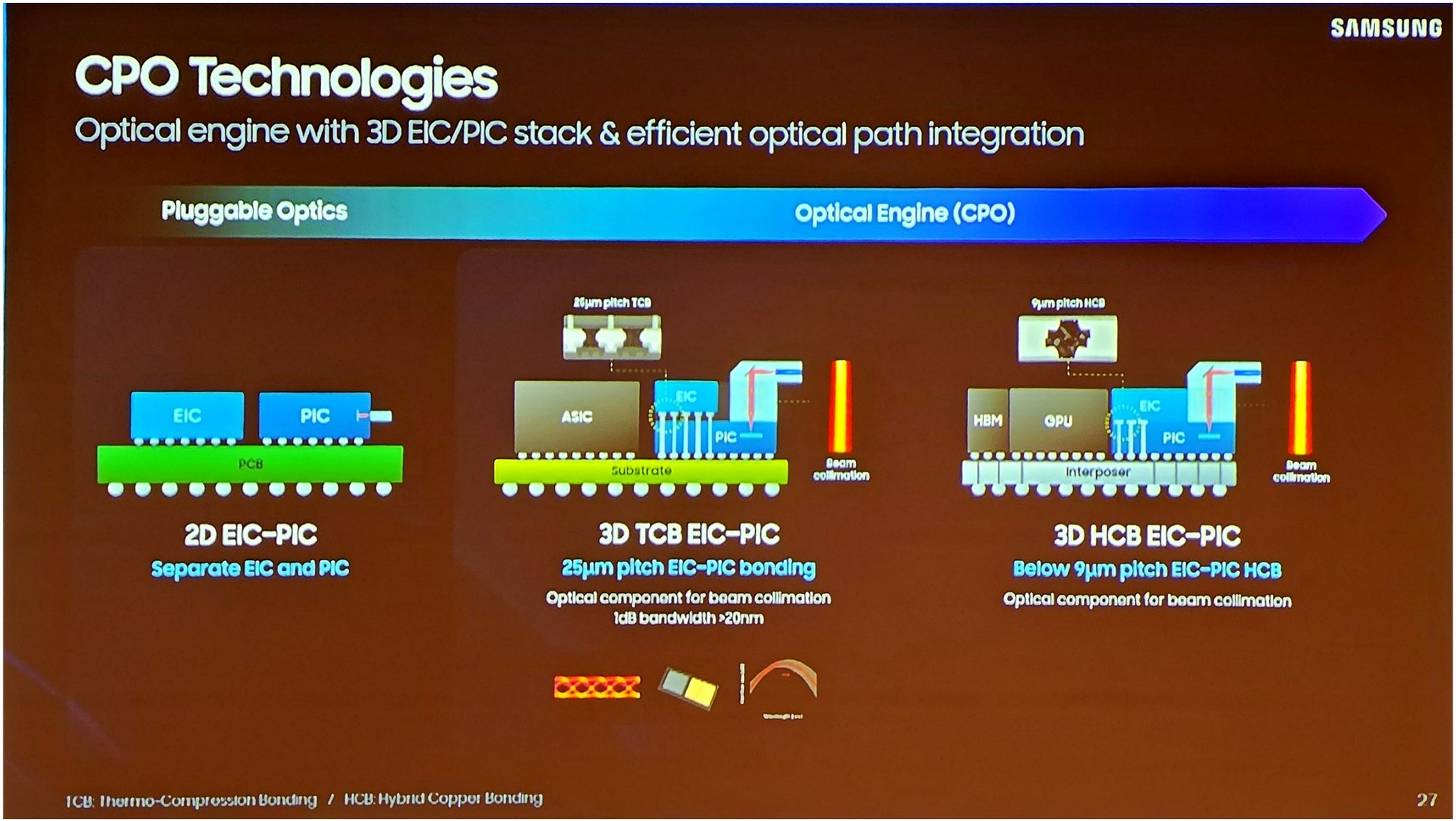
Co-packaged Optics · 9W

No DSP → Power saving



	Photonics	Interconnect	Optical Engine	ASIC-Optics Distance	Optical Engine (OE) PKG Type
ENERGY PER BIT 20 pJ/bit	PO	Rack to Rack	External	100–300mm+	2D PKG
	XPO	Rack to Rack	External (w/ CPC)	100–300mm+	3D Stack PKG
	NPO	Rack to Rack	Internal on Board	25–75mm	
<5 pJ/bit	CPO Switch	Rack to Rack	Internal on PKG Substrate	2–5mm	
	CPO XPU	XPU PKG to PKG	Internal on PKG Substrate or Interposer	0.5–2mm	

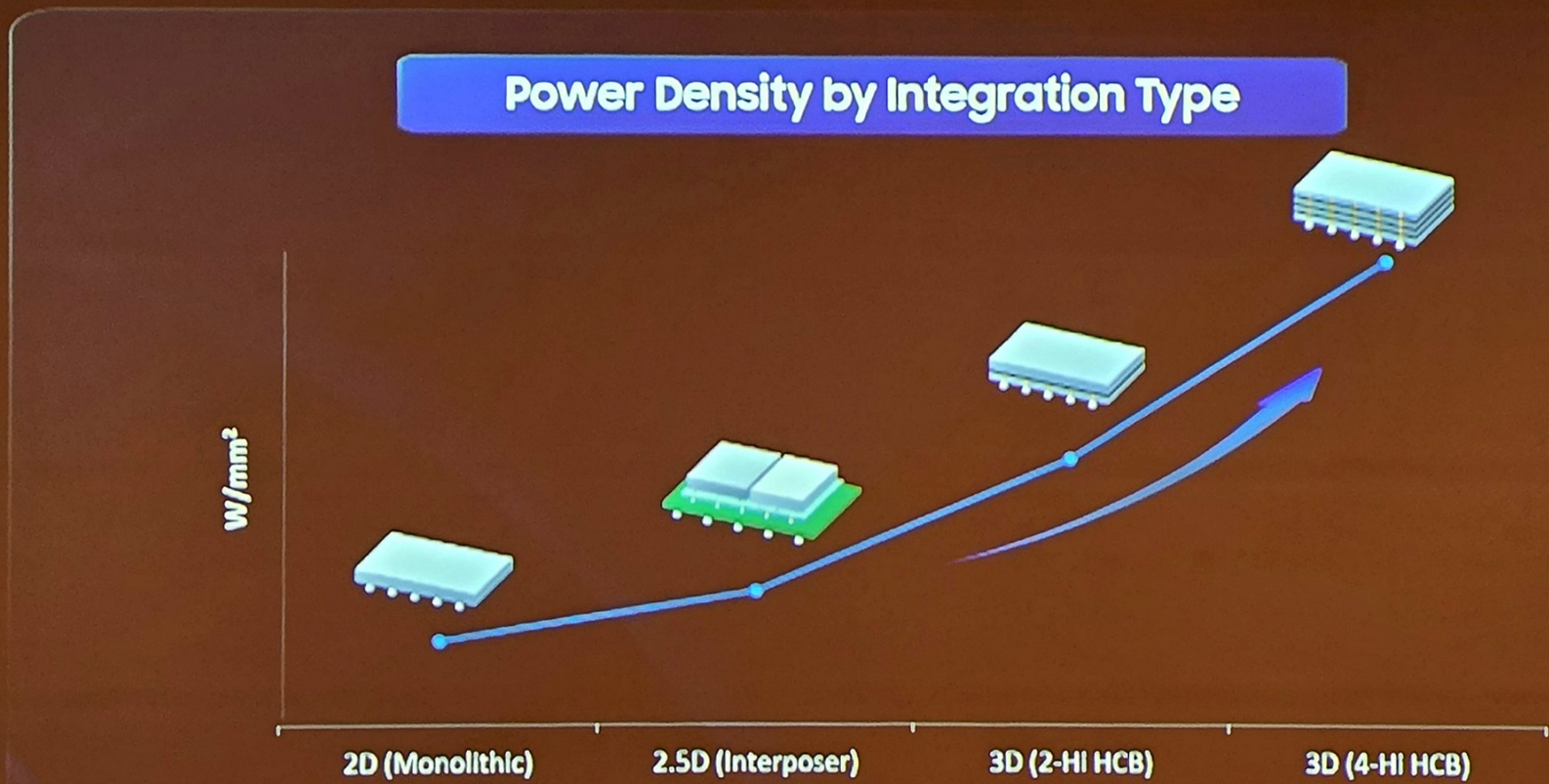
26


SAMSUNG
CPO Technologies
Optical engine with 3D EIC/PIC stack & efficient optical path integration
Pluggable Optics
Optical Engine (CPO)
EIC
PIC
PCB
2D EIC-PIC
Separate EIC and PIC
25μm pitch TCB
ASIC
EIC
PIC
Substrate
Beam collimation
3D TCB EIC-PIC
25μm pitch EIC-PIC bonding
Optical component for beam collimation
1dB bandwidth >20nm
9μm pitch HCB
HBM
GPU
EIC
PIC
Interposer
Beam collimation
3D HCB EIC-PIC
Below 9μm pitch EIC-PIC HCB
Optical component for beam collimation
TCB: Thermo-Compression Bonding / HCB: Hybrid Copper Bonding
27

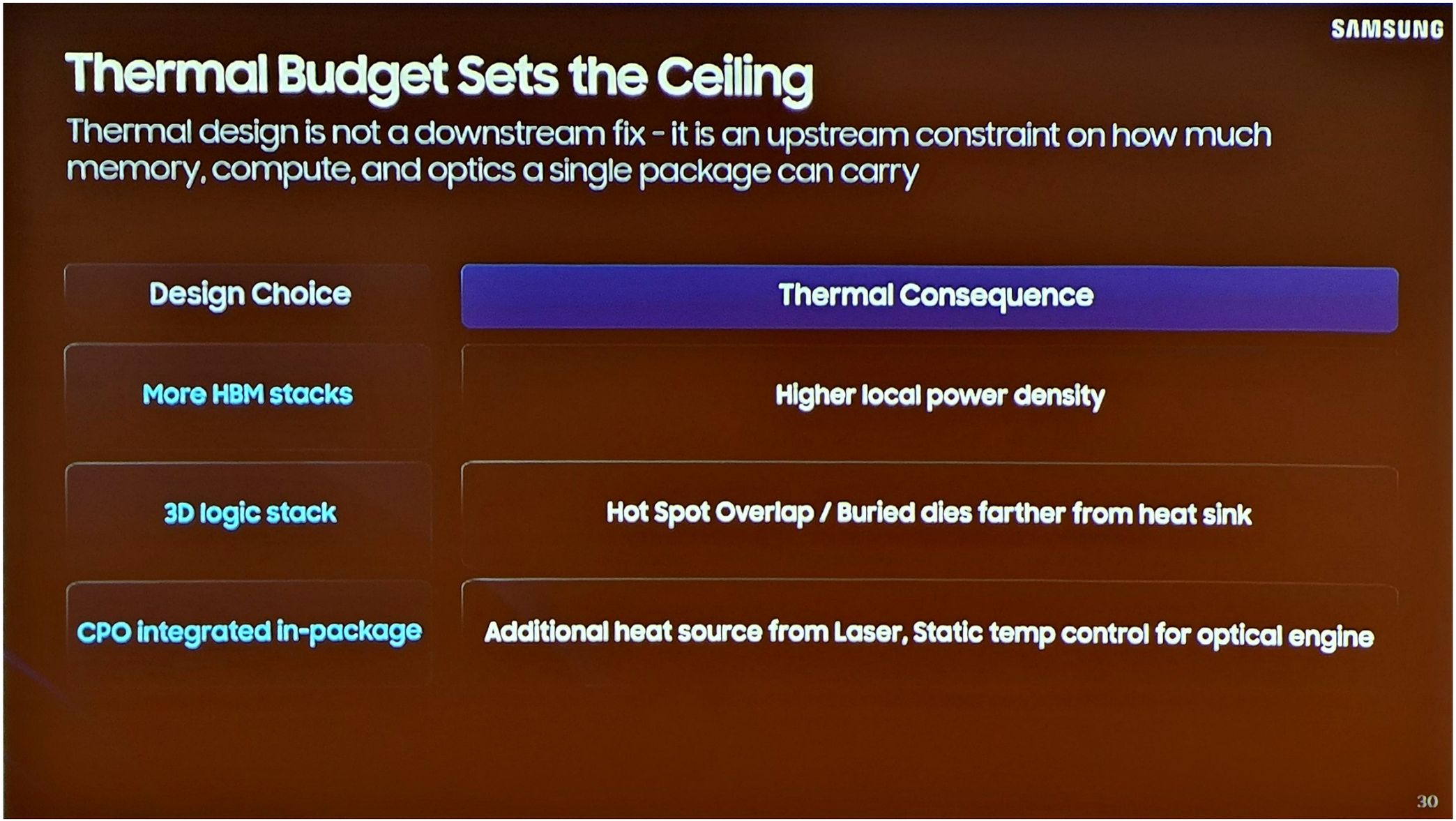
SAMSUNG

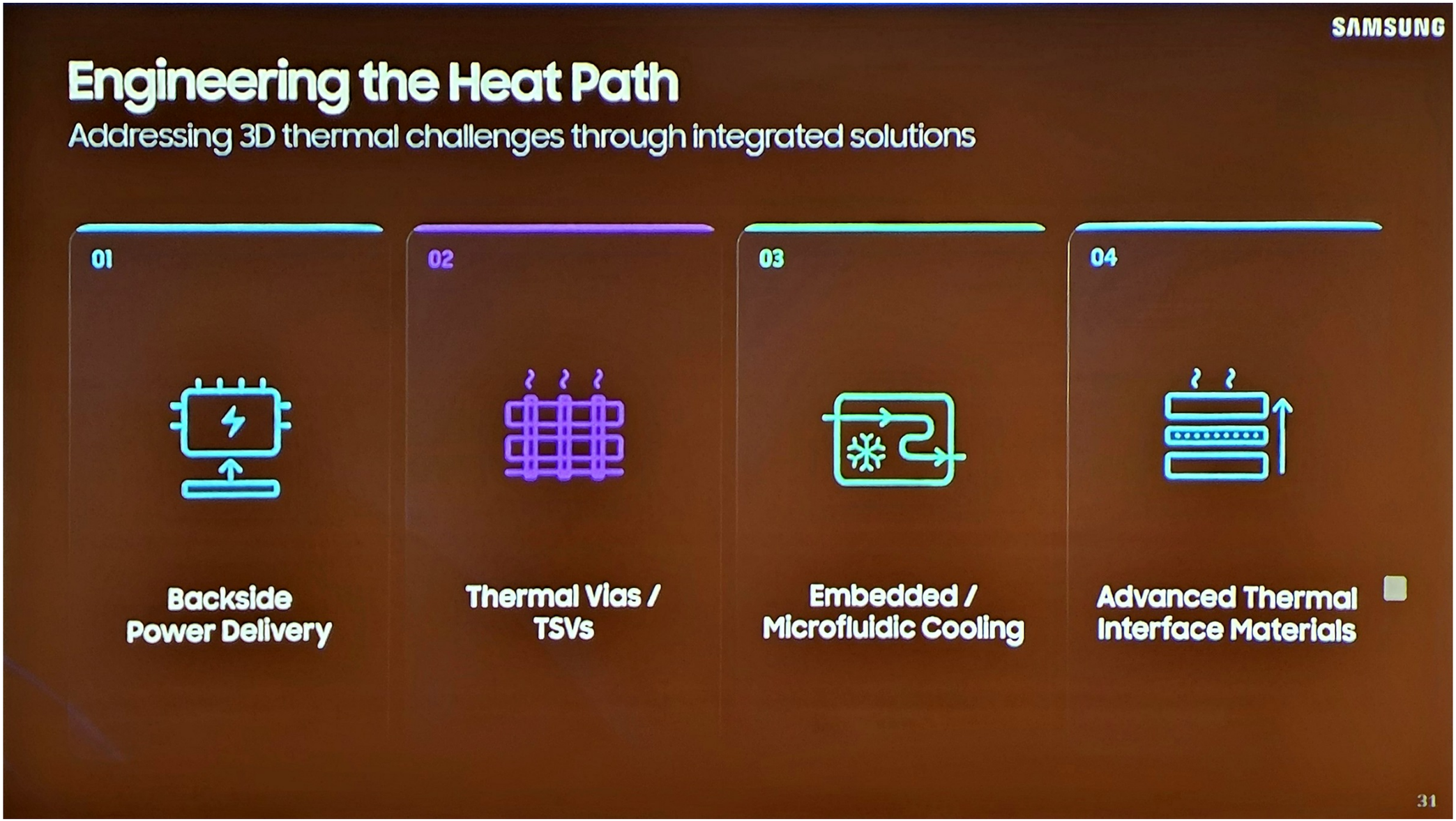
Thermal Management: The Hidden Constraint

Stacking logic on logic multiplies power density in the same footprint
Every gain from 3D HCB creates a heat-removal problem that did not exist in 2D



29

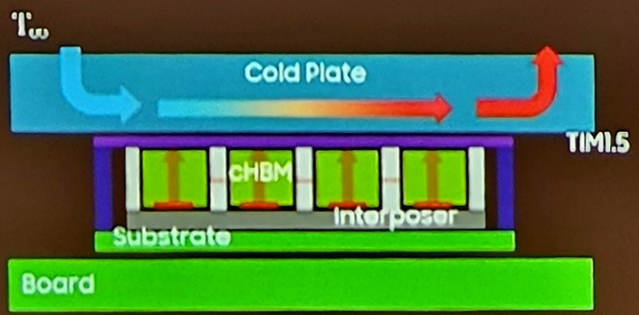

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Thermal Budget Sets the Ceiling
Thermal design is not a downstream fix - it is an upstream constraint on how much memory, compute, and optics a single package can carry
Design Choice
Thermal Consequence
More HBM stacks
Higher local power density
3D logic stack
Hot Spot Overlap / Buried dies farther from heat sink
CPO integrated in-package
Additional heat source from Laser, Static temp control for optical engine
30


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Engineering the Heat Path
Addressing 3D thermal challenges through integrated solutions
01
Backside Power Delivery
02
Thermal Vias / TSVs
03
Embedded / Microfluidic Cooling
04
Advanced Thermal Interface Materials
31

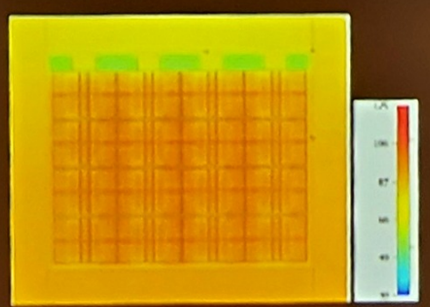
SAMSUNG

Thermal Solution for Advanced Packages

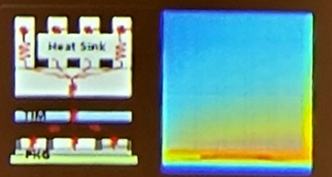
PKG design reducing thermal crosstalk and thermal resistance



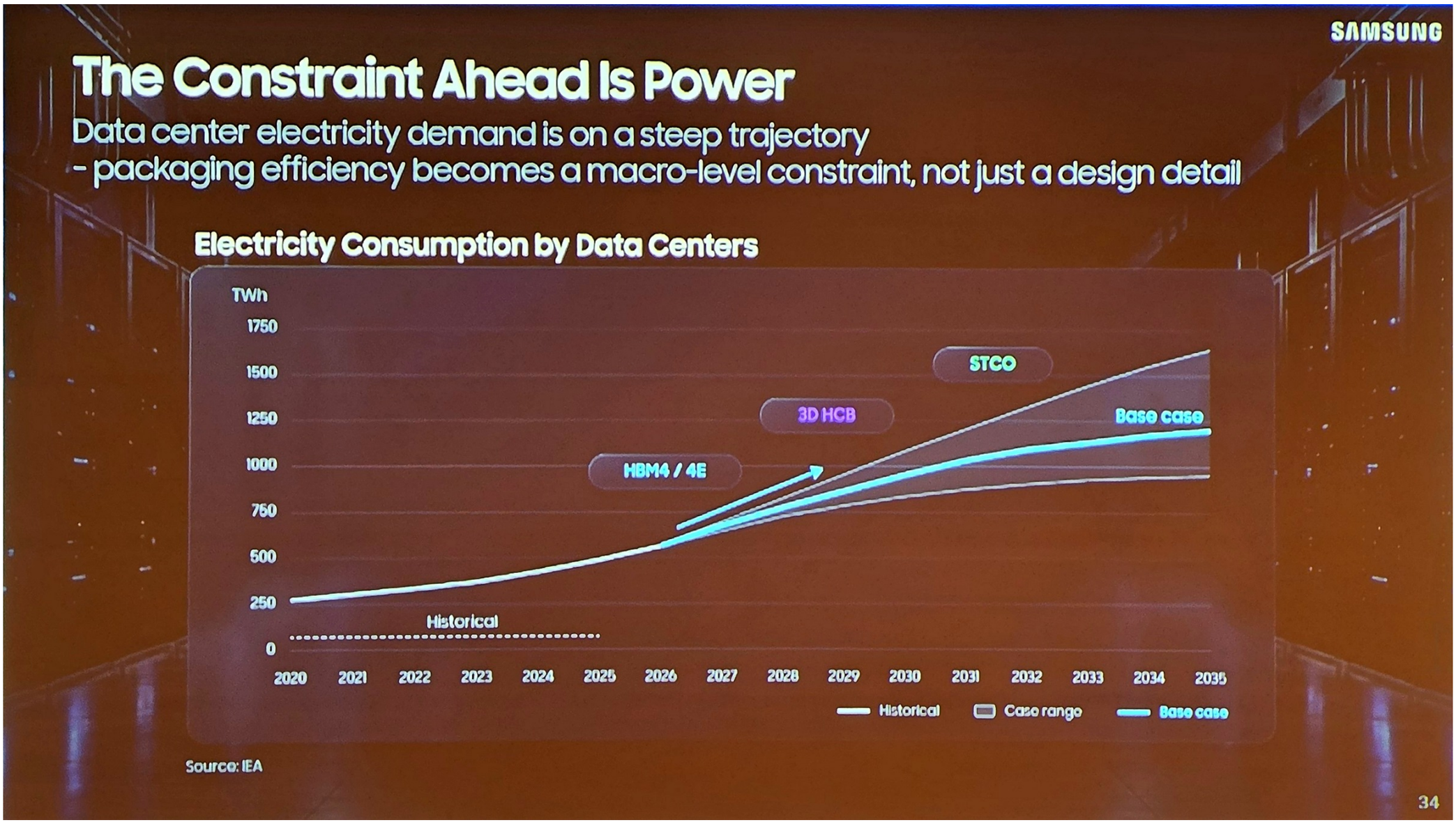
[Liquid cooling system environment]

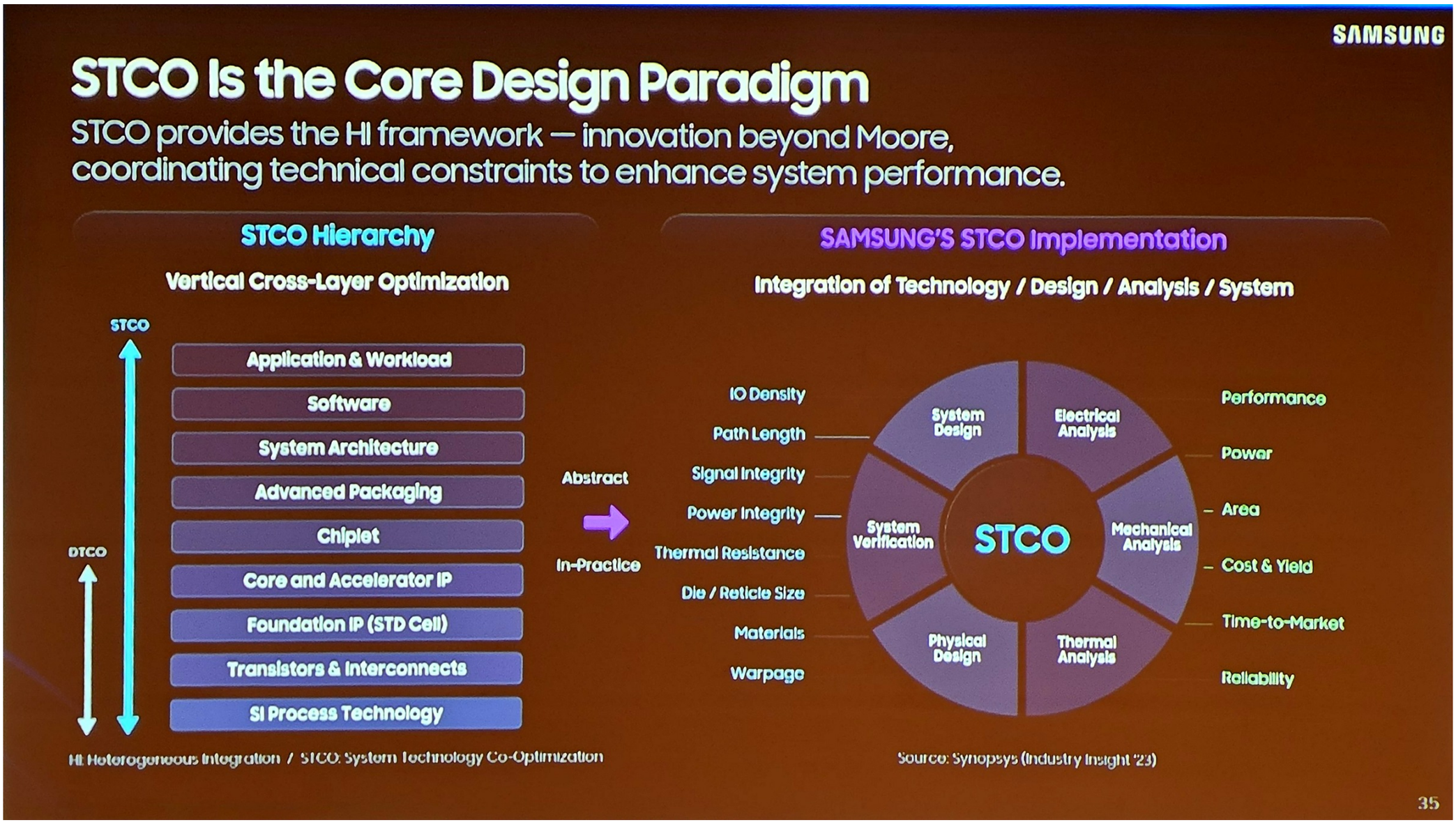


[Temperature contour for SIP]

Category \ Domain	Chip-PKG Interconnection level	PKG level	PKG-system Interconnection level
In-plane			
Through-plane			
Thermal crosstalk	IP Block ↔ IP Block	Chip ↔ Chip Chip ↔ PKG	PKG ↔ PKG
Main mechanism	Conduction	Conduction, Convection	Convection, Radiation

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The Constraint Ahead Is Power
Data center electricity demand is on a steep trajectory
- packaging efficiency becomes a macro-level constraint, not just a design detail
Electricity Consumption by Data Centers
TWh
1750
1500
1250
1000
750
500
250
0
STCO
3D HCB
Base case
HBM4 / 4E
Historical
2020
2021
2022
2023
2024
2025
2026
2027
2028
2029
2030
2031
2032
2033
2034
2035
Historical
Case range
Base case
Source: IEA
34


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STCO Is the Core Design Paradigm
STCO provides the HI framework — innovation beyond Moore,
coordinating technical constraints to enhance system performance.
STCO Hierarchy
Vertical Cross-Layer Optimization
STCO
DTCO
Application & Workload
Software
System Architecture
Advanced Packaging
Chiplet
Core and Accelerator IP
Foundation IP (STD Cell)
Transistors & Interconnects
SI Process Technology
HI: Heterogeneous Integration / STCO: System Technology Co-Optimization
SAMSUNG'S STCO Implementation
Integration of Technology / Design / Analysis / System
Abstract
In-Practice
IO Density
Path Length
Signal Integrity
Power Integrity
Thermal Resistance
Die / Reticle Size
Materials
Warpage
System Design
Electrical Analysis
System Verification
STCO
Mechanical Analysis
Physical Design
Thermal Analysis
Performance
Power
Area
Cost & Yield
Time-to-Market
Reliability
Source: Synopsys (Industry Insight '23)
35

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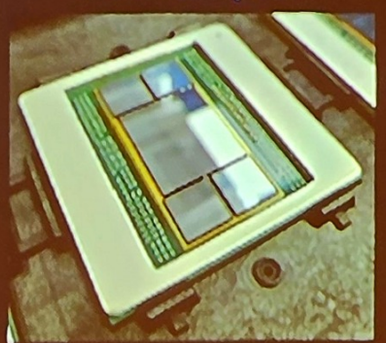
2.xD Solutions for AI Data Center

Bandwidth and scalability achieved through 2.xD integration

AI Solution for Data Center

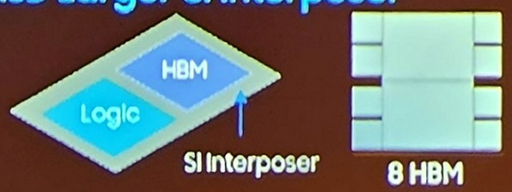
Today's Industry AI Chip

Silicon Interposer



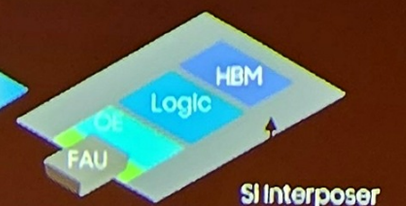
Bandwidth: 3.3TB/s
Capacity: 96GB

2.5D Larger Si Interposer



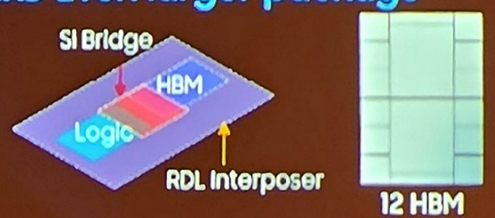
2.5D Cube S

Bandwidth: 2x @ 6.6 TB/s
Capacity: 2x @ 192 GB



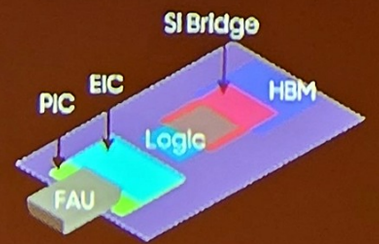
Interposer Solution Expansion to CPO

2.xD Even larger package

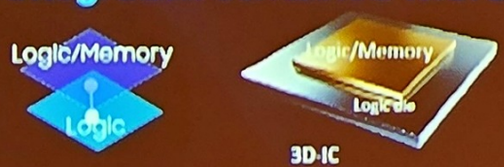


2.3D Cube E

Bandwidth: 9x @ 30.7 TB/s
Capacity: 6x @ 576 GB



3D Integration combined with 2.xD



3D Vertical Integration

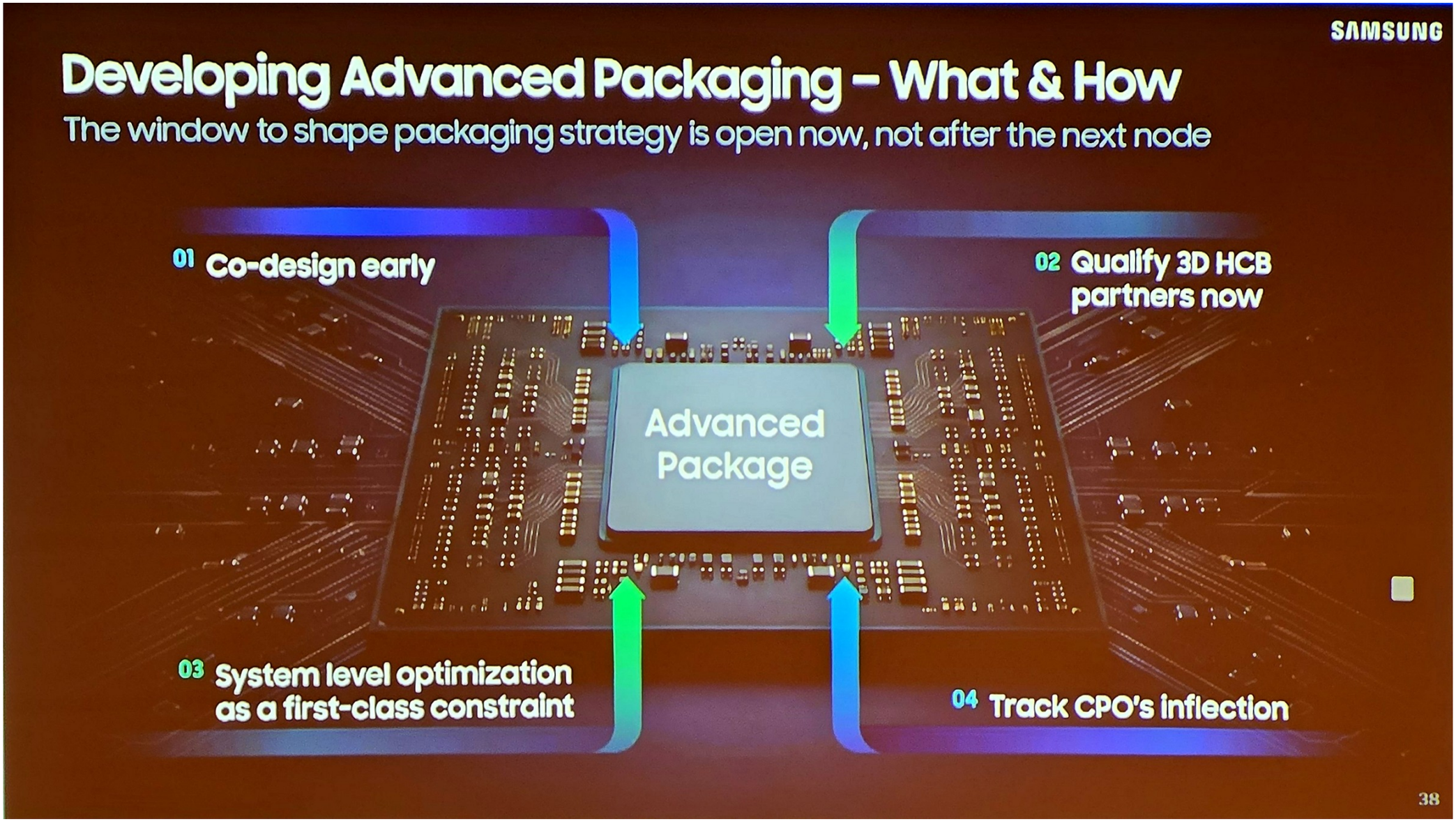
Logic/Logic or Logic/Memory
Bandwidth : 21x @ 70.5 TB/s

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T18 메모리 로직 광학을 연결하는 패키징 공동설계

Samsung Electronics · Won Kyoung Choi




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Developing Advanced Packaging – What & How
The window to shape packaging strategy is open now, not after the next node
01 Co-design early
02 Qualify 3D HCB partners now
Advanced Package
03 System level optimization as a first-class constraint
04 Track CPO's inflection
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